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Some Basic Concepts of Chemistry - Mole Concept

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1. The volume of a gas at 0°C and 760 mm pressure is 22.4 cc. The no. of molecules present in this volume is
(1) 10^{-3} N_A (2) 10^{-4} N_A
(3) 10^{-5} N_A (4) 10^{-2} N_A
CMC174

2. 100 mL of PH_3 when completely decomposed produces phosphorus and hydrogen. The change in volume of the gas is -

$$PH_3(g) \longrightarrow 2P(s) + \frac{3}{2}H_2(g)$$

(1) 50 mL increase (2) 500 mL decrease
(3) 900 mL decrease (4) No change
CMC175

3. The volume of CO_2 at STP obtained by heating 1 g of $CaCO_3$ will be
(1) 1 L (2) 22.4 L
(3) 0.224 L (4) 11.2 L
CMC176

4. 3 L of mixture of propane (C_3H_8) & butane (C_4H_{10}) on complete combustion gives10 L CO_2. Find the composition of mixture.
(1) C_3H_8, 2L and C_4H_{10}, 1L
(2) C_3H_8, 3L and C_4H_{10}, 0L
(3) C_3H_8, 1.5L and C_4H_{10}, 1.5L
(4) C_3H_8, 0L and C_4H_{10}, 3L
CMC177

5. The mass of a molecule of water :
(1) 30×10^{-24} kg (2) 3×10^{-26} kg
(3) 1.5×10^{-26} kg (4) 2.5×10^{-27} kg
CMC178

6. At NTP, 5.6 L of a gas weigh 8 g. The vapour density of gas is :-
(1) 32 (2) 40
(3) 16 (4) 8
CMC179

7. 0.01 mole of iodoform (CHI_3) reacts with Ag to produce a gas whose volume at NTP is

$$2CHI_3 + 6Ag \longrightarrow 6AgI(s) + C_2H_2(g)$$

(1) 224 mL (2) 112 mL
(3) 336 mL (4) None of these
CMC180

8. 19.7 kg of gold was recovered from a smuggler. How many atoms of gold were recovered ? (At. wt. of gold (Au) = 197)
(1) 100 (2) 6.023×10^{23}
(3) 6.023×10^{24} (4) 6.023×10^{25}
CMC181

9. Chlorine is prepared in the laboratory by treating manganese dioxide (MnO_2) with aqueous hydrochloric acid according to the reaction

$$4HCl(aq) + MnO_2(s) \rightarrow 2H_2O(\ell) + MnCl_2(aq) + Cl_2(g)$$

How many gram of HCl react with 5.0 g of manganese dioxide ? (At. wt. of Mn = 55)
(1) 2.12 g (2) 44.24 g
(3) 8.4 g (4) 3.65 g
CMC182

10. 510 milli g of a liquid on vapourisation produces 510 cc at NTP. The molecular weight of liquid is -
(1) 0.224 (2) 2.24
(3) 224 (4) 22.4
CMC183

11. An unknown amino acid has 0.032% sulphur by mass. If each molecule has one sulphur atom only, 1 g of amino acid has ..... molecules :-
(1) 6.02×10^{18} (2) 6.02×10^{19}
(3) 6.02×10^{24} (4) 6.02×10^{23}
CMC184

12. 446 g of PbO, 46 g of NO_2 and 16 g of O_2 are allowed to react according to the equation :-

$$PbO + 2NO_2 + \frac{1}{2}O_2 \rightarrow Pb(NO_3)_2$$

The amount of $Pb(NO_3)_2$ that can be produced is (At. wt. of Pb = 207) :-
(1) 331 g (2) 662 g
(3) 165.5 g (4) None of these
CMC185

13. A sample of organic compound on complete combustion gives 88 g CO_2 and 36 g of H_2O. The molecular formula of the compound may be :-
(1) C_4H_6 (2) CH_4
(3) C_2H_4 (4) C_2H_6
CMC186

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14. Volume at NTP of 0.22 g of CO_2 is same as that of
(1) 0.01 g of hydrogen
(2) 0.085 g of NH_3
(3) 320 mg of gaseous SO_2
(4) All of the above.

CMC187

15. The weight of 1×10^{22} molecules of $CuSO_4.5H_2O$ is (At. wt. of Cu = 63.5)
(1) 41.44 g (2) 414.5 g
(3) 4.144 g (4) 4145 g

CMC188

16. A hydrocarbon contain 80%C. The vapour density of compound in 30. Molecular formula of compound is :-
(1) CH_3 (2) C_2H_6
(3) C_4H_{12} (4) All of these

CMC189

17. Which of the following has maximum mass :-
(1) 0.1 **gram atom** of carbon
(2) 0.1 mol of ammonia
(3) 6.02×10^{22} molecules of hydrogen
(4) 1120 cc of carbon dioxide at STP

CMC190

18. What mass of NaCl would be decomposed by 98 g of H_2SO_4 if 120 g of $NaHSO_4$ and 27.5 g of HCl are produced in a reaction. Assuming that law of mass conservation is true:
(1) 4.95 g (2) 49.5 g
(3) 0.495 g (4) 495 g

CMC191

19. Equal mass of oxygen, hydrogen and methane are taken in a container in identical conditions. What is the ratio of their mole :-
(1) 1 : 16 : 1 (2) 1 : 16 : 2
(3) 8 : 1 : 8 (4) 16 : 1 : 8

CMC192

20. One litre of a certain gas weighs 1.16 g at STP. The gas may be -
(1) C_2H_2 (2) CO
(3) O_2 (4) NH_3

CMC193

21. When 40 cc of slightly moist hydrogen chloride gas is mixed with 20 cc of ammonia gas the final volume of gas left at the same temperature and pressure will be
$NH_3(g) + HCl(g) \longrightarrow NH_4Cl(s)$
(1) 20 cc (2) 40 cc
(3) 60 cc (4) 100 cc

CMC194

22. The vapour densities of two gases are in the ratio of 1 : 3. Their molecular masses are in the ratio of :-
(1) 1 : 3 (2) 1 : 2
(3) 2 : 3 (4) 3 : 1

CMC195

23. Calculate the number of atoms present in 7.1 g of chlorine?
(1) 0.2 N_A (2) 0.3 N_A
(3) 0.4 N_A (4) 0.5 N_A

CMC196

24. For reaction A + 2 B $\longrightarrow$ C. The amount of C formed by starting the reaction with 5 mole of A and 8 mole of B is :
(1) 5 mol (2) 8 mol
(3) 16 mol (4) 4 mol

CMC197

25. In a compound, x is 75.8% and y is 24.2% by weight present. If atomic weight of x and y are 24 and 16 respectively. Then calculate the empirical formula of the compound.
(1) x_2y (2) xy_2
(3) xy (4) x_3y_3

CMC198

26. At 25°C for complete combustion of 5 mol propane (C_3H_8). The required volume of O_2 at STP will be.
(1) 560 L (2) 560 mL
(3) 360 L (4) 360 mL

CMC199

27. 1.5 moles of O_2 combines with Mg to form oxide MgO. The mass of Mg (At. mass 24) that has combined is $2Mg + O_2 \longrightarrow 2MgO$
(1) 72 g (2) 36 g
(3) 24 g (4) 94 g

CMC200

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28. At STP, for complete combustion of 3 g C_2H_6 the required volume of O_2 will be -
(1) 78.4 L (2) 7.84 L
(3) 2.78 L (4) 6.23 L
CMC201

29. 0.6 g of carbon was burnt in the air to form CO_2. The number of molecules of CO_2 introduced into the air will be :
$C + O_2 \longrightarrow CO_2$
(1) 6.02×10^{23} (2) 3.01×10^{23}
(3) 6.02×10^{22} (4) 3.01×10^{22}
CMC202

30. 5 L of gas at STP weighs 6.25 g. What is the gram molecular weight ?
(1) 1.25 (2) 14 (3) 28 (4) 56
CMC203

31. Two elements X (At wt = 75) and Y (At wt = 16) combine to give a compound having 75.8% of X. The empirical formula of compound is :
(1) XY (2) X_2Y (3) X_2Y_2 (4) X_2Y_3
CMC204

32. The density of air is 0.001293 g mL^{-1} at NTP. Its vapour density is :-
(1) 143 (2) 14.53
(3) 1.43 (4) 0.143
CMC205

33. One mol of CO_2 contains –
(1) 3 atom
(2) 18.1×10^{23} molecules of CO_2
(3) 6.02×10^{23} atom of oxygen
(4) 6.02×10^{23} atom of carbon
CMC206

34. Which one of the following pairs of gases contains the same number of molecules ?
(1) 16 g of O_2 and 14 g of N_2
(2) 8 g of O_2 and 22 g of CO_2
(3) 28 g of N_2 and 22 g of CO_2
(4) 32 g of O_2 and 32 g of N_2
CMC207

35. The number of oxygen atoms in 4.4 g of CO_2 is approx.
(1) 1.2×10^{23} (2) 6×10^{22}
(3) 6×10^{23} (4) 12×10^{23}
CMC208

36. 1 mole of CH_4 contains :
(1) 6.02×10^{23} atoms of H
(2) 4 g atom of hydrogen
(3) 1.81×10^{23} molecules of CH_4
(4) 3.0 g of carbon.
CMC209

37. Calculate the number of atom present in one drop of water having mass 1.8 g.
(1) 0.3 N_A (2) 0.2 N_A
(3) 0.1 N_A (4) 0.01 N_A
CMC210

38. In the reaction, $4A + 2B + 3C \longrightarrow A_4B_2C_3$ what will be the number of moles of product formed, starting from 1 mole of A, 0.6 mole of B and 0.72 mole of C :-
(1) 0.25 (2) 0.3 (3) 2.4 (4) 0.24
CMC211

39. In an organic compound of molar mass 108 g mol^{-1} C, H and N atoms are present in 9 : 1 : 3.5 by weight. Molecular formula can be:-
(1) $C_6H_8N_2$ (2) $C_7H_{10}N$
(3) $C_5H_6N_3$ (4) $C_4H_{18}N_3$
CMC212

40. 44g of a sample of organic compound on complete combustion gives 88g CO_2 and 36g of H_2O. The molecular formula of the compound may be :-
(1) C_4H_6 (2) C_2H_6O
(3) C_2H_4O (4) C_3H_6O
CMC213

41. Which of the following has greatest number of atoms :-
(1) 1g of butane (C_4H_{10})
(2) 1g of nitrogen (N_2)
(3) 1g of silver (Ag)
(4) 1g of water (H_2O)
CMC214

42. One atom of an element 'X' weighs 6.664×10^{-23} g. The number of gram atoms in 40 kg of it is
(1) 10 (2) 100 (3) 10000 (4) 1000
CMC215

43. If $Ca_3(PO_4)_2$ and H_3PO_3 contain same number of 'P' atom then the ratio of oxygen atoms in these compounds respectively is :-
(1) 8/3 (2) 2/3 (3) 3 (4) 4/3
CMC216

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44. Calculate the number of oxygen atoms required to combine with 7g of N_2 to form N_2O_3 when 80% of N_2 is converted into N_2O_3:-
(1) 2.3×10^{23} (2) 3.6×10^{23}
(3) 1.8×10^{21} (4) 5.4×10^{21}
CMC217

45. Element X reacts with oxygen to produce a pure sample of X_2O_3. In an experiment it is found that 1.00 g of X produces 1.16 g of X_2O_3. Calculate the atomic weight of X :-
(1) 67 (2) 100.2 (3) 125 (4) 150
CMC218

46. In a compound A containing carbon, hydrogen and oxygen, the % of hydrogen is one sixth that of carbon which is 1.5 times that of oxygen. What is the simplest formula for the compound:-
(1) $C_2H_4O_2$ (2) C_2H_4O
(3) C_2H_6O (4) CH_4O
CMC219

47. 1 mole of Fe_2S_3, 2 moles of H_2O and 3 moles of O_2 are allowed to react according to the equation
$[2Fe_2S_3(s) + 6H_2O(\ell) + 3O_2(g) \rightarrow 4Fe(OH)_3(s) + 6S(s)]$
The number of moles of $Fe(OH)_3(s)$ that can be produced is :-
(1) 1.34 (2) 2 (3) 4 (4) 3
CMC220

48. A gaseous alkane was burnt with oxygen. The volume of O_2 for complete combustion and that of CO_2 formed are in the ratio of 7 : 4. The molecular formula of alkane is :-
(1) CH_4 (2) C_2H_6
(3) C_3H_8 (4) C_4H_{10}
CMC221

49. Two oxides of a metal contain 22.22% and 30% oxygen by mass respectively. If the formula of the first oxide is MO, then the formula of the second oxide is :-
(1) MO (2) MO_2 (3) M_2O_3 (4) M_2O
CMC222

50. Caffeine has a molecular weight of 194. If it contains 29% by mass of nitrogen, number of atoms of nitrogen in one molecule of caffeine is :-
(1) 2 (2) 6 (3) 4 (4) 3
CMC223

51. Sulphur trioxide is prepared by the following two reactions.
$S_8(s) + 8O_2(g) \rightarrow 8SO_2(g)$
$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$
How many grams of SO_3 are produced from 1 mol of S_8 ?
(1) 1280.0 (2) 640.0
(3) 960.0 (4) 320.0
CMC224

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	3	1	2	3	2	4	3	4	1	3	3	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	2	2	1	1	1	1	4	1	1	1	2	4	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	2	4	1	1	2	1	4	1	3	1	4	4	2	4
Question	46	47	48	49	50	51									
Answer	2	1	2	3	3	2									

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Atomic Structure

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1. The triad of nuclei that is isotonic :-

(1) ${}^{14}_{6}C$, ${}^{15}_{7}N$, ${}^{17}_{9}F$ (2) ${}^{14}_{6}C$, ${}^{14}_{7}N$, ${}^{17}_{9}F$

(3) ${}^{13}_{6}C$, ${}^{14}_{7}N$, ${}^{19}_{9}F$ (4) ${}^{14}_{6}C$, ${}^{14}_{7}N$, ${}^{19}_{9}F$

CAS231

2. X^+, Y^{2+} and Z^- are isoelectronic with CO_2. Increasing order of protons in X^+, Y^{2+} and Z^- is:

(1) $X^+ = Y^{2+} = Z^-$ (2) $X^+ < Y^{2+} < Z^-$

(3) $Z^- < X^+ < Y^{2+}$ (4) $Y^{2+} < X^+ < Z^-$

CAS232

3. The number of photons emitted in 10 hours by a 60 W sodium lamp λ of photon = 6000 Å

(1) 6.50×10^{24} (2) 6.40×10^{23}

(3) 8.40×10^{23} (4) 3.40×10^{23}

CAS233

4. The energy needed to excite an electron of hydrogen atom from its ground state to its third excited state is :-

(1) 12.1 eV (2) 10.2 eV

(3) 0.85 eV (4) 12.75 eV

CAS234

5. In which of the following transitions will the wavelength be minimum ?

(1) n = 6 to n = 4 (2) n = 4 to n = 2

(3) n = 3 to n = 1 (4) n = 2 to n = 1

CAS235

6. Hydrogen atom electron is excited to n = 4 state, in the spectrum of emitted radiation, number of lines in the ultravoilet and visible regions are respectively:-

(1) 3, 1 (2) 1, 3 (3) 2, 3 (4) 3, 2

CAS236

7. In which of the following pairs is the probability of finding the electron in only xy - plane is zero for both orbitals ?

(1) $3d_{yz}$, $4d_{x^2-y^2}$ (2) $2P_z$, $3d_{z^2}$

(3) $4d_{zx}$, $3P_z$ (4) $2P_z$, $3P_z$

CAS237

8. If the total energy of an electron in a hydrogen atom in an excited state is –3.4 eV, then the de-broglie wavelength of the electron is :-

(1) 7.2×10^{-10} m (2) 3×10^{-10} m

(3) 5×10^{-9} m (4) 9.3×10^{-12} m

CAS238

9. Magnetic moment of X^{+3} ion of 3d series is $\sqrt{35}$ B.M. What is the atomic number of X^{+3} ?

(1) 24 (2) 25 (3) 26 (4) 27

CAS239

10. Which orbit would be the first to have 'h' subshell:-

(1) 3rd (2) 4th

(3) 5th (4) 6th

CAS240

11. What are the values of the orbital angular momentum of an electron in the orbitals 1s, 3s, 3d, and 2p :-

(1) 0, 0, $\sqrt{6}\hbar$, $\sqrt{2}\hbar$ (2) 1, 1, $\sqrt{6}\hbar$, $\sqrt{2}\hbar$

(3) 0, 1, $\sqrt{6}\hbar$, $\sqrt{3}\hbar$ (4) 0, 0, $\sqrt{20}\hbar$, $\sqrt{6}\hbar$

CAS241

12. The energy corresponding to second line of Balmer series for hydrogen atom will be :-

(1) 12.1 eV (2) 1.89 eV

(3) 2.55 eV (4) 13.6 eV

CAS242

13. If the velocity of an electron in Ist orbit of H is V, then what will be the velocity in 4th orbit of Be^{+3} :-

(1) V (2) $\frac{V}{4}$

(3) 16 V (4) 4 V

CAS243

14. The wavelength of the radiation emitted when an electron falls from Bohr's orbit 4 to 2 in H atom is

(1) 972 nm (2) 486 nm

(3) 243 nm (4) 182 nm

CAS244

15. According to Bohr's atomic model

(1) Electron in an atom of hydrogen can have only certain values of angular momentum

(2) Electrons have particle as well as wave character

(3) Atomic spectrum of an element should contain only five lines

(4) All the above statements are incorrect

CAS245

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16. Which of the d-orbitals lies in the xy-plane ?
(1) d_{xy} only (2) d_{xz} only
(3) d_{z^2} only (4) d_{xy} and d_{z^2}
CAS246

17. The line with smallest wavelength in the Balmer series in the hydrogen spectrum will have the frequency :-
(1) $8.22 \times 10^{14}\ s^{-1}$
(2) $3.29 \times 10^{15}\ s^{-1}$
(3) $7.22 \times 10^{16}\ s^{-1}$
(4) $8.05 \times 10^{13}\ s^{-1}$
CAS247

18. The two particles A and B have de Broglie wavelengths 1 nm and 5 nm respectively. If mass of A is four times the mass of B, the ratio of kinetic energies of A and B would be:-
(1) 5 : 1 (2) 25 : 4
(3) 20 : 1 (4) 5 : 4
CAS248

19. The specific charge of a proton is $9.6 \times 10^7 C\ kg^{-1}$, then for an α-particle it will be -
(1) $2.4 \times 10^7 C\ kg^{-1}$
(2) $4.8 \times 10^7 C\ kg^{-1}$
(3) $19.2 \times 10^7 C\ kg^{-1}$
(4) $38.4 \times 10^7 C\ kg^{-1}$
CAS249

20. It is known that an atom contains protons, neutrons and almost electrons. If the mass of neutron is assumed to half of its original value where as that of proton is assumed to be twice of its original value then the atomic mass of $^{14}_{7}N$ will be :-
(1) same (2) 25 % more
(3) 14.28 % more (4) 28.5 % less
CAS250

21. In hydrogen atom the energy difference between the states n = 2 and n = 3 is E eV. The ionisation potential of H atom is -
(1) 3.2 E (2) 5.6 E
(3) 7.2 E (4) 13.2 E
CAS251

22. Which of the following statement (regarding an atom of H) is correct –
(1) Kinetic energy of the electron is maximum in the first orbit
(2) Potential energy of the electron is maximum in the first orbit
(3) Radius of the second orbit is two times the radius of the first orbit
(4) Various energy levels are equally spaced
CAS252

23. When the electron of a hydrogen atom jumps from the n = 4 to the n = 1 state, the number of all possible spectral lines emitted is :-
(1) 15 (2) 6 (3) 3 (4) 4
CAS253

24. An electron in an atom jumps in such a way that its kinetic energy changes from x to $\frac{x}{4}$. The change in potential energy will be :-
(1) $+\frac{3}{2}x$ (2) $-\frac{3}{8}x$ (3) $+\frac{3}{4}x$ (4) $-\frac{3}{4}x$
CAS254

25. Which orbital notation does not have spherical node :-
(1) n = 2, ℓ = 0 (2) n = 3, ℓ = 1
(3) n = 3, ℓ = 0 (4) n = 1, ℓ = 0
CAS255

26. In how many elements does the last electron have the quantum number of n = 4 and ℓ = 1?
(1) 4 (2) 6 (3) 8 (4) 10
CAS256

27. For principle quantum number n = 4, the total number of orbitals having ℓ = 3 is :-
(1) 3 (2) 7 (3) 5 (4) 9
CAS257

28. Choose the wrong one on the basis of bohr theory :-
(1) Radius of an orbit $\propto \frac{n^2}{z}$
(2) Velocity of $e^- \propto \frac{z}{n}$
(3) Energy of e^- in an orbit $\propto \frac{n^2}{z^2}$
(4) mvr $\propto$ n
CAS258

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29. In which transition minimum energy is emitted

(1) $\infty \rightarrow 1$ (2) $2 \rightarrow 1$

(3) $3 \rightarrow 2$ (4) $n \rightarrow (n-1)$ $[n \geq 4]$

CAS259

30. If the value of principal quantum number is 4, the possible values for magnetic quantum number will be :-

(1) 4 (2) 16 (3) 32 (4) 9

CAS260

31. If the value of E = –78.4 kCal / mol, the serial no. of the orbit in hydrogen atom is :-

(1) 4 (2) 3 (3) 2 (4) 1

CAS261

32. The most intense line in the Brackett series of the spectrum of atomic hydrogen is the transition :-

(1) $n_2 = \infty \rightarrow n_1 = 1$ (2) $n_2 = 8 \rightarrow n_1 = 4$

(3) $n_2 = \infty \rightarrow n_1 = 4$ (4) $n_2 = 4 \rightarrow n_1 = 3$

CAS262

33. Five valence electrons of ^{15}P are labelled as



If the spin quantum numbers of B and Z is $+\frac{1}{2}$, the group of electron with three of the quantum number (n, *l*, m) same are.

(1) AB, XYZ, BY (2) AB

(3) XYZ, AZ (4) AB, XYZ

CAS263

34. The I.P. of an atom is 16 eV. What is the value of first excitation potential :–

(1) 10·2 eV (2) 12 eV

(3) 14 eV (4) 16 eV

CAS264

35. Distance travelled by e^- during one revolution in an orbit in H atom is :-

(1) Inversely proportional to shell no.

(2) Directly proportional to shell no.

(3) Directly proportional to square of shell no.

(4) Inversely proportional to square of shell no.

CAS265

36. An electron is present in 4f sub-shell. The possible values of azimuthal quantum number for this electron are :-

(1) 0, 1, 2, 3 (2) 1, 2, 3, 4

(3) 3 (4) 4

CAS266

37. An atom has 35 nucleons and has atomic number equal to 17. The number of electrons with n = 2, m = 0 in it is :-

(1) 2 (2) 4 (3) 6 (4) 3

CAS267

38. A dye absorbs a photon of wavelength λ and re-emits the same energy into two photons of wavelengths λ_1 and λ_2 respectively. The wavelength λ is related with λ_1 and λ_2 as :-

(1) $\lambda = \frac{\lambda_1 + \lambda_2}{\lambda_1 \lambda_2}$ (2) $\lambda = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$

(3) $\lambda = \frac{\lambda_1^2 \lambda_2^2}{\lambda_1 + \lambda_2}$ (4) $\lambda = \frac{\lambda_1 \lambda_2}{(\lambda_1 + \lambda_2)^2}$

CAS268

39. The number of photons of light having wave number 'x' in 10 J of energy source is :-

(1) 10hcx (2) $\frac{hc}{10x}$

(3) $\frac{10}{hcx}$ (4) None of these

CAS269

40. Energy required for the process is :

$He^+(g) \longrightarrow He^{+2}(g) + e^-$

The ionisation energy for the H-atom in the ground state is 2.18×10^{-18} J atom^{-1}

(1) 9.12×10^{-18} J (2) 8.72×10^{-18} J

(3) 4.72×10^{-20} J (4) 7.82×10^{-16} J

CAS270

41. When an electron of mass 6m and charge 6e moves with a velocity of 7v around the nuclear charge Ze in a circular orbit of radius 'r'. The potential energy of the electron will be:-

(1) $-\frac{Ze^2}{r}$ (2) $+\frac{Ze^2}{r}$

(3) $-\frac{6Ze^2}{r}$ (4) $-\frac{2\pi^2 me^4 Z^2}{n^2 h^2}$

CAS271

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42. A single electron in an ion has ionisation energy equal to 217.6 eV. What is the total number of neutrons present in that ion is :-
(1) 2 (2) 4 (3) 5 (4) 9
CAS272

43. How fast is an electron moving if it has a wavelength equal to the distance it travels in one second ?
(1) $\sqrt{\frac{h}{m}}$ (2) $\sqrt{\frac{m}{h}}$
(3) $\sqrt{\frac{h}{P}}$ (4) $\sqrt{\frac{h}{2(KE)}}$
CAS273

44. The difference in angular momentum associated with the electron in two successive orbits of hydrogen atom is :-
(1) $\frac{h}{\pi}$ (2) $\frac{h}{2\pi}$
(3) $\frac{h}{2}$ (4) (n–1)
CAS274

45. If uncertainties in the measurement of position and momentum are equal, uncertainty in the measurement of velocity.
(1) $\frac{1}{m}\sqrt{\frac{h}{\pi}}$ (2) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$
(3) $\sqrt{\frac{h}{\pi}}$ (4) None of these
CAS275

46. Find out the number of waves made by a Bohr electron in one complete revolution in its fourth orbit.
(1) 3 (2) 4 (3) 2 (4) 1
CAS276

47. If a 1.00 g body is travelling along the x-axis at 100 cm s^{-1} within 1 cm s^{-1}, what is the theoretical uncertainty in its position (in meter)
(1) 5.25×10^{-30} (2) 6×10^{-10}
(3) 10×10^{-8} (4) 6.42×10^{-28}
CAS277

48. The amount of energy required to remove electron from a Li^{+2} ion in its ground state is how many times greater than the amount of energy needed to remove the electron from an H atom in its ground state ?
(1) 9 (2) 6 (3) 4 (4) 5
CAS278

49. What is the difference in wavelengths of the 4th and 5th lines of the Balmer series in the spectrum of atomic hydrogen :-
(1) 131 Å (2) 520 Å
(3) 390 Å (4) 262 Å
CAS279

50. In a certain electronic transition in the hydrogen atom from an initial state (i) to a final state (f) the difference in the orbit radius ($r_i - r_f$) is seven times the Bohr radius. Identify the transition :-
(1) $4 \rightarrow 1$ (2) $4 \rightarrow 2$
(3) $4 \rightarrow 3$ (4) $3 \rightarrow 1$
CAS280

51. A hydrogen atom in an excited state is ionised by the absorption of radiation with a photon energy of 1 eV. What is the minimum value of n for excited state of the atom :-
(1) 1 (2) 2 (3) 3 (4) 4
CAS281

52. Which of the following statements regarding an orbital is correct :-
(1) An orbital is a definite trajectory around the nucleus in which electron can move
(2) An orbital always has spherical trajectory
(3) It is the region around the nucleus where there is 90% to 95% probability of finding all electrons in an atom
(4) An orbital is characterised by the three distinct quantum numbers n, ℓ and m
CAS282

53. Following transitions occuring in hydrogen atoms ($n_1 < n_2 < n_3$)
n_1 orbit $\xrightarrow{1}$ n_2 orbit
n_2 orbit $\xrightarrow{2}$ n_3 orbit
n_1 orbit $\xrightarrow{3}$ n_3 orbit
Which of the following relation is not correct?
(1) $\lambda_3 = \lambda_1 + \lambda_2$ (2) $\overline{\nu}_3 = \overline{\nu}_1 + \overline{\nu}_2$
(3) $\nu_3 = \nu_1 + \nu_2$ (4) $\Delta E_3 = \Delta E_1 + \Delta E_2$
CAS283

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54. In Cr atom, the number of electrons having n + m = 3 are :-
(1) 5 (2) 6 (3) 7 (4) 8
CAS284

55. Two particles A and B are in motion, wavelength related to particle A is 5×10^{-8} m. If the momentum of particle B is half of A then what will be the wavelength of B :-
(1) 2.5×10^{-8} m (2) 1.25×10^{-8} m
(3) 1×10^{-7} m (4) 1×10^{-8} m
CAS285

56. The value of $(n_2 + n_1)$ and $(n_2^2 - n_1^2)$ for $He^{\oplus}$ ion in atomic spectrum are 4 and 8 respectively. The wavelength of emitted photon on transition of electron from n_2 to n_1 is :-
(1) $\frac{32\ R_H}{9}$ (2) $\frac{9\ R_H}{32}$
(3) $\frac{9}{32\ R_H}$ (4) $\frac{32}{9\ R_H}$
CAS286

57. If the ionization energy of He^+ is 19.6×10^{-18} J $atom^{-1}$, then calculate the energy of first stationary state of Li^{2+}.
(1) -1.1×10^{-17} J/atom
(2) -1.1×10^{-18} J/atom
(3) -4.41×10^{-17} J/atom
(4) -4.41×10^{-18} J/atom
CAS287

58. What is the frequency of a photon emitted during a transition from n = 4 state to the n = 2 state in the hydrogen atom ?
(1) 6.15×10^{11} Hz (2) 6.15×10^{14} Hz
(3) 3.12×10^{7} Hz (4) 1.23×10^{9} Hz
CAS288

59. A moving electron has 4.55×10^{-25} Joules of kinetic energy. Calculate its wavelength ($m_e = 9.1 \times 10^{-31}$ Kg)
(1) 2.55×10^{-6} m (2) 3.57×10^{-5} m
(3) 1.45×10^{-4} m (4) 7.25×10^{-7} m
CAS289

60. The angular momentum of an electron is a Bohr's orbit of H atom is 4.2178×10^{-34} kg-m^2/sec. Calculate the wavelength of spectral line emitted when electron falls from this level to next lower level.
(1) 1.8×10^{-4} cm (2) 2.8×10^{-5} cm
(3) 4.6×10^{-4} cm (4) 2.2×10^{-6} cm
CAS290

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	3	1	4	3	4	4	1	3	4	1	3	1	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	1	2	2	2	3	1	2	1	4	2	2	3	4	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	2	2	3	3	2	2	3	2	3	3	1	2	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	1	1	3	4	4	1	3	3	3	3	2	4	1

Redox Reactions

1. When ferrous oxalate is titrated against $K_2Cr_2O_7$, milli equivalents of Fe^{2+}, $C_2O_4^{2-}$ and $Cr_2O_7^{2-}$ in the redox reaction are X, Y and Z respectively, then:-

(1) X = Y (2) X + Y = Z

(3) X + 2Y = Z (4) 2X + 6Y = 6Z

CRR138

2. The equivalent mass of Mn in the following change is $Mn \longrightarrow Mn^{2+} + Mn^{7+}$

(1) $\frac{55}{9}$ (2) $\frac{55}{2}$

(3) $\frac{55}{7}$ (4) $\frac{55\times2}{9}$

CRR139

3. What mass of MnO_2 is reduced by 35 mL of 0.16 N oxalic acid ($H_2C_2O_4$) in acidic solution? The skeleton equation is :-

$MnO_2 + H^+ + H_2C_2O_4 \rightarrow CO_2 + H_2O + Mn^{2+}$

(1) 8.7 g (2) 0.24 g

(3) 0.84 g (4) 43.5 g

CRR140

4. How many gram of $KMnO_4$ are contained in 4 litre of 0.05 N solution. The $KMnO_4$ is to be used as an oxidant in acidic medium ?

(1) 1.58 g (2) 15.8 g

(3) 6.32 g (4) 31.6 g

CRR141

5. A compound contains element X, Y, Z with Oxidation no. +3, +5 and –2 respectively. What will be possible formula of compound containing above element.

(1) XYZ (2) $Y_2(XZ_3)_2$

(3) $X_3(YZ_4)_3$ (4) $X_3(Y_4Z)_2$

CRR142

6. Which of the following can only acts as oxidising agent?

(1) $KMnO_4$ (2) K_2MnO_4

(3) H_2O_2 (3) SO_2

CRR143

7. Balance the following reaction

$aZn + NO_3^- + bH_2O \xrightarrow{OH^-} cZn^{2+} + NH_4^+ + dOH^-$

(1) a = 4, b = 7, c = 4, d = 10

(2) a = 2, b = 5, c = 6, d = 8

(3) a = 3, b = 3, c = 1, d = 9

(4) a = 1, b = 3, c = 7, d = 8

CRR144

8. How many mole of $FeSO_4$, $H_2C_2O_4$ and FeC_2O_4 are oxidised separately by one mole of $KMnO_4$ in acidic medium respectively :-

(1) 5 ; 5/2 ; 5/3 (2) 2/5 ; 2 ; 5

(3) 5/3 ; 3/5 ; 2/5 (4) 3/5 ; 2 ; 5

CRR145

9. Which of the following reagent can act as reducing agent with SO_2 :-

(1) Cl_2 (2) $KMnO_4$

(3) HNO_3 (4) H_2S

CRR146

10. Which of the following is not a disproportionation reaction :-

(1) $P_4 + NaOH \rightarrow NaH_2PO_2 + PH_3$

(2) $BaC_2 + N_2 \rightarrow Ba(CN)_2$

(3) $Hg_2I_2 \rightarrow HgI_2 + Hg$

(4) $Cl_{2(g)} + 2OH^-_{(aq)} \rightarrow ClO^-_{(aq)} + Cl^-_{(aq)} + H_2O$

CRR147

11. Which of the following is correctly matched :-

Species	**Oxidation number**
(a) $\underline{Al}F_3$	(P) +2.5
(b) $\underline{P}_4O_{10}$	(Q) +7
(c) $H\underline{Cl}O_4$	(R) +5
(d) $\underline{S}_4O_6^{2-}$	(S) +3

	(a)	**(b)**	**(c)**	**(d)**
(1)	P	Q	R	S
(2)	S	R	Q	P
(3)	S	R	P	Q
(4)	P	Q	S	R

CRR148

12. When BrO_3^- ion reacts with Br^- in acidic medium, Br_2 is liberated. The equivalent weight of Br_2 in this reaction is :-

(1) $\frac{5M}{8}$ (2) $\frac{5M}{3}$ (3) $\frac{3M}{5}$ (4) $\frac{4M}{6}$

CRR149

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13. $N_2 + 3H_2 \longrightarrow 2NH_3$ molecular weight of NH_3 and N_2 are x_1 and x_2 their equivalent weight are y_1 and y_2. Then $(y_1 - y_2)$ is :-

(1) $\left(\frac{2x_1 - x_2}{6}\right)$ (2) $(x_1 - x_2)$

(3) $(3x_1 - x_2)$ (4) $(x_1 - 3x_2)$

CRR150

14. Oxidation number of all three Fe atom in Fe_3O_4 is :-

(1) 3, 3, 3 (2) 2, 2, 2

(3) 2, 2, 3 (4) 2, 3, 3

CRR151

15. Which reaction does not represent auto redox or disproportionation ?

(1) $Cl_2 + OH^- \longrightarrow Cl^- + CLO_3^- + H_2O$

(2) $2H_2O_2 \longrightarrow H_2O + O_2$

(3) $2Cu^+ \longrightarrow Cu^{2+} + Cu$

(4) $(NH_4)_2Cr_2O_7 \longrightarrow N_2 + Cr_2O_3 + 4H_2O$

CRR152

16. The oxidation number of oxygen in KO_3, Na_2O_2 is :-

(1) 3, 2 (2) 1, 0

(3) 0, 1 (4) –0.33, –1

CRR153

17. Which species are oxidized and reduced in the reaction ?

$FeC_2O_4 + KMnO_4 \longrightarrow Fe^{3+} + CO_2 + Mn^{2+}$

(1) Oxidised : Fe, C ; Reduced : Mn

(2) Oxidised : Fe ; Reduced : Mn

(3) Reduced : Fe, Mn ; Oxidised : C

(4) Reduced : C ; Oxidised : Mn, Fe

CRR154

18. Equivalent weight of FeS_2 in the half reaction, $FeS_2 \longrightarrow Fe_2O_3 + SO_2$ is :-

(1) M/10 (2) M/11 (3) M/6 (4) M/1

CRR155

19. In the balanced chemical reaction,

$IO_3^- + aI^- + bH^+ \longrightarrow cH_2O + dI_2$

a, b, c and d respectively correspond to :-

(1) 5, 6, 3, 3 (2) 5, 3, 6, 3

(3) 3, 5, 3, 6 (4) 5, 6, 5, 5

CRR156

20. Match incorrect equivalent mass of reactant in following column ? (M = molar mass of reactant):-

Column-I	Column-II
(1) $FeS_2 \rightarrow Fe^{3+} + SO_3$	$\frac{M}{15}$
(2) $P_2H_4 \rightarrow P_4H_2 + PH_3$	$\frac{5M}{6}$
(3) $2Mn^{+7} \rightarrow Mn^{2+} + Mn^{+6}$	$\frac{M}{3}$
(4) $Ba(MnO_4)_2 \xrightarrow{H^+} Mn^{2+}$	$\frac{M}{5}$

CRR157

21. An element 'A' in a compound ABD has oxidation number –n. It is oxidised by $Cr_2O_7^{2-}$ in acidic medium. In the experiment 1.68×10^{-3} moles of $K_2Cr_2O_7$ were used for 3.36×10^{-3} moles of ABD. The new oxidation number of A after oxidation is :-

(1) 3 (2) 3 – n (3) n – 3 (4) + n

CRR158

22. In the following redox reaction :-

$Cu_3P + Cr_2O_7^{2-} \longrightarrow Cu^{+2} + H_3PO_4 + Cr^{+3}$

equivalent weight of Cu_3P is

(1) $\frac{M}{3}$ (2) $\frac{M}{11}$ (3) $\frac{M}{8}$ (4) $\frac{M}{13}$

CRR159

23. Which of the following is not disproportionation

(1) $P_4 + 5OH^- \rightarrow H_2PO_4^- + PH_3$

(2) $Cl_2 + OH^- \rightarrow Cl^- + ClO^-$

(3) $2H_2O_2 \rightarrow 2H_2O + O_2$

(4) $PbO_2 + H_2O \rightarrow PbO + H_2O_2$

CRR160

24. Which of the following is intermolecular redox reaction :-

(1) $2\begin{matrix}CHO\\|\\CHO\end{matrix} \xrightarrow{OH^-} \begin{matrix}CH_2OH\\|\\COOH\end{matrix}$

(2) $2C_6H_5CHO \xrightarrow{Al(OC_2H_5)_3} C_6H_5COOH + C_6H_5CH_2OH$

(3) $4CrO_5 + 6H_2SO_4 \rightarrow 2Cr_2(SO_4)_3 + 6H_2O + 7O_2$

(4) $As_2S_3 + HNO_3 \rightarrow H_3AsO_4 + H_2SO_4 + NO$

CRR161

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25. In the redox reaction

$xMnO + yPbO_2 + zHNO_3 \rightarrow$

$HMnO_4 + Pb(NO_3)_2 + H_2O$

(1) x = 2, y = 5, z = 10 (2) x = 2, y = 7, z = 8

(3) x = 2, y = 5, z = 8 (4) x = 2, y = 5, z = 5

CRR162

26. Hydrazine reacts with KIO_3 in presence of HCl as; $N_2H_4 + IO_3^- + 2H^+ + Cl^- \rightarrow ICl + N_2 + 3H_2O$

The equivalent masses of N_2H_4 and KIO_3 respectively are : (K = 39, I = 127) :-

(1) 8,87 (2) 8,35.6

(3) 16,53.5 (4) 8,53.5

CRR163

27. 12.5 mL of 0.05 M SeO_2 oxidises 25 mL of 0.1 M $CrSO_4$ solution to $Cr_2(SO_4)_3$. What is the final oxidation number of Se ?

(1) +2 (2) +4

(3) +6 (4) 0

CRR164

28. What is the equivalent mass of As_2S_3 in the following reaction : (Molar mass of As_2S_3 = M)

$As_2S_3 + NO_3^- + H^+ \rightarrow As_2O_5 + NO_2^- + SO_3 + H_2O$

(1) M/4 (2) M/24

(3) M/28 (4) M/12

CRR165

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	2	3	3	1	1	1	4	2	2	3	1	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	4	1	2	1	4	2	2	4	4	1	4	4	3		

Thermodynamics

1. Given the following entropy values (in $JK^{-1}\ mol^{-1}$) at 298 K and 1 atm: $H_2(g)$:130.6, $Cl_2(g)$: 223.0 and HCl(g) : 186.7. The entropy change (in $JK^{-1}\ mol^{-1}$) for the reaction

$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$ is

(1) + 540.3 (2) 727.0

(3) – 166.9 (4) 19.8

CTD274

2. Standard heat of combustion of C(s), $H_2(g)$ and $CH_4(g)$ are – 94, – 68 and – 213 kCal/mol. then $\Delta H°$ for $C(s) + 2H_2(g) \rightarrow CH_4(g)$ is :-

(1) – 17 kCal (2) – 111 kCal

(3) – 170 kCal (4) – 85 kCal

CTD275

3. Considering entropy (S) as a thermodynamic parameter, the criteria for the spontaneity of any process is :-

(1) $\Delta S_{system} - \Delta S_{surroundings} > 0$

(2) $\Delta S_{system} > 0$ only

(3) $\Delta S_{surroundings} > 0$ only

(4) $\Delta S_{system} + \Delta S_{surroundings} > 0$

CTD276

4. Which one of the following has $\Delta S°$ greater than zero :-

(1) $CaO(s) + CO_2(g) \rightleftharpoons CaCO_3(s)$

(2) $NaCl(aq) \rightleftharpoons NaCl(s)$

(3) $NaNO_3(s) \rightleftharpoons Na^+(aq) + NO_3^-(aq)$

(4) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

CTD277

5. The bond dissociation energies of H_2, Cl_2 and HCl are 104, 58 and 103 kCal respectively. The enthalpy of formation of HCl is :-

(1) + 44 kCal (2) + 22 kCal

(3) – 22 kCal (4) – 44 kCal

CTD278

6. The internal energy change when a system goes from state A to B is 40 kJ/mol. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy :

(1) < 40 kJ (2) Zero

(3) 40 kJ (4) > 40 kJ

CTD279

7. If at 298 K the bond energies of C – H, C – C, C = C and H – H bonds are respectively 414, 347, 615 and 435 kJ mol^{-1}, the value of enthalpy change for the reaction

$H_2C = CH_2(g) + H_2(g) \rightarrow H_3C - CH_3(g)$ at 298K will be :–

(1) + 125 kJ (2) – 125 kJ

(3) + 250 kJ (4) – 250 kJ

CTD280

8. The free energy change for the following reactions are given below

$C_2H_2(g) + \frac{5}{2} O_2(g) \rightarrow 2CO_2(g) + H_2O(\ell)$;

$\Delta G° = -1234$ kJ

$C(s) + O_2(g) \rightarrow CO_2(g)$; $\Delta G° = -394$ kJ

$H_2(g) + \frac{1}{2} O_2(g) \rightarrow H_2O(\ell)$; $\Delta G° = -237$ kJ

What is the standard free energy change for the reaction $H_2(g) + 2C(s) \rightarrow C_2H_2(g)$:–

(1) – 209 kJ (2) – 2259 kJ

(3) + 2259 kJ (4) 209 kJ

CTD281

9. The ΔH_f^0 for $CO_2(g)$, CO(g) and $H_2O(g)$ –393.5, –110.5 and –241.8kJ mol^{-1} respectively. The standard enthalpy change (in kJ) for the reaction $CO_2(g) + H_2(g) \rightarrow CO(g) + H_2O(g)$ is

(1) 524.1 (2) 41.2

(3) –262.5 (4) – 41.2

CTD282

10. Heat required to raise the temperature of 1 mol of a gas of a substance by 1°C is called :-

(1) Specific heat

(2) Molar heat capacity

(3) Enthalpy

(4) Internal energy

CTD283

11. Which is intensive property :-

(1) Mass (2) Mass/volume

(3) Volume (4) None

CTD284

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12. Which is correct about ΔG :-
(1) $\Delta G = \Delta H - T\Delta S$
(2) At equilibrium $\Delta G° = 0$
(3) At equilibrium $\Delta G = -RT \log_e k$
(4) $\Delta G = \Delta G° + RT \log_e k$

CTD285

13. ΔE = 0, for which process:-
(1) Cyclic process
(2) Isobaric process
(3) Ioschoric process
(4) Adiabatic process

CTD286

14. Which of the following statements are true :-
(1) Work is a state function
(2) Temperature is a state function
(3) Entropy is path function
(4) Heat is a state function

CTD287

15. Which of the following is true for spontaneous process :-
(1) $\Delta G > 0$ (2) $\Delta G < 0$
(3) $\Delta G = 0$ (4) $\Delta G = T\Delta S$

CTD288

16. In which reaction ΔS is negative :-
(1) $H_2O(\ell) \rightarrow H_2O(g)$
(2) $H_2O(s) \rightarrow H_2O(\ell)$
(3) $H_2O(g) \rightarrow H_2O(s)$
(4) $H_2O(s) \rightarrow H_2O(g)$

CTD289

17. Which of the following value of heat of formation indicates that the product is least stable
(1) –100 kcal (2) –200 kcal
(3) +20 kcal (4) +60 kcal

CTD290

18. Enthalpy change for reaction
$1/2\ H_2(g) + 1/2Cl_2(g) \rightarrow HCl(g)$ is called:-
(1) Enthalpy of combustion
(2) Enthalpy of solution
(3) Enthalpy of formation
(4) Enthalpy of fusion

CTD291

19. For the reaction $PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$
(1) $\Delta H = \Delta E$ (2) $\Delta H > \Delta E$
(3) $\Delta H < \Delta E$ (4) None

CTD292

20. In an adiabatic expansion of an ideal gas-
(1) $W = -\Delta E$ (2) $W = \Delta E$
(3) $\Delta E = 0$ (4) $W = 0$

CTD293

21. The heat of combustion of ethanol in a bomb calorimeter is –670.48 kcal mol^{-1} at 25°C. What is ΔE at 25°C for the reaction ?
(1) – 269.24 kcal (2) – 469.28 kcal
(3) – 670.48 kcal (4) +770.48 kcal

CTD294

22. $4S(s) + 6O_2 \longrightarrow 4SO_3(g)$.; Δ H for this reaction is – 1583.2 kJ. The enthalpy of formation of sulphur trioxide is -
(1) – 395.8 kJ (2) 395.8 kJ
(3) 495.5 kJ (4) 595.5 kJ

CTD295

23. The heat change for the reaction
$C(s) + 2S(s) \longrightarrow CS_2(\ell)$ is called the heat of
(1) Reaction
(2) Fusion
(3) Formation
(4) Combustion

CTD296

24. Based on the following thermochemical equation
$H_2O(g) + C(s) = CO(g) + H_2(g)$; Δ H = 131 kJ,
$CO(g) + ½\ O_2(g) = CO_2(g)$; Δ H = –282 kJ
$H_2(g) + ½\ O_2(g) = H_2O(g)$; Δ H = –242 kJ
$C(s) + O_2(g) = CO_2(g)$; Δ H = – x kJ
The value of x will be
(1) 393 kJ (2) – 393 kJ
(3) 655 kJ (4) – 655 kJ

CTD297

25. In a reversible reaction of the type
$A + B \rightleftharpoons AB$, in general
(1) The combination reaction will be exothermic and the dissociation reaction will be endothermic
(2) The combination reaction will be endothermic and the dissociation reaction will be exothermic
(3) Both the reaction will be exothermic
(4) Neither of the reactions will be endothermic

CTD298

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26. If enthalpies of formation for $C_2H_4(g)$, $CO_2(g)$ and $H_2O(\ell)$ at 25°C and 1 atm, pressure are 52, –394 and –286 kJ/mol respectively, then enthalpy of combustion of $C_2H_4(g)$ will be
(1) – 141.2 kJ/mol (2) –1412 kJ/mol
(3) 14.2 kJ/mol (4) +14.12 kJ/mol
CTD299

27. Given that $C + O_2 \longrightarrow CO_2$, $\Delta H^0 = -x$ kJ,
$2CO + O_2 \longrightarrow 2CO_2$; $\Delta H^0 = -y$ kJ
The enthalpy of formation of carbon monoxide will be
(1) y – 2x (2) $\frac{2x-y}{2}$
(3) $\frac{y-2x}{2}$ (4) 2x – y
CTD300

28. Identify the correct statement regarding entropy.
(1) At absolute zero temperature, the entropy of perfectly crystalline substance is +ve.
(2) At absolute zero temp. entropy of perfectly crystalline substance is taken to be zero.
(3) At 0°C the entropy of a perfectly crystalline substance is taken to be zero.
(4) At absolute zero temperature, the entropy of all crystalline substances is taken to be zero.
CTD301

29. Change in enthalpy for reaction
$2H_2O_2(\ell) \rightarrow 2H_2O(\ell) + O_2(g)$
If heat of formation of $H_2O_2(\ell)$ and $H_2O(\ell)$ are –188 & – 286 kJ/mol respectively :–
(1) – 196 kJ/mol (2) + 196 kJ/mol
(3) + 948 kJ/mol (4) – 948 kJ/mol
CTD302

30. When 1 mol gas is heated at constant volume temperature is raised from 298 to 308 K. Heat supplied to the gas is 500 J. Then which statement is correct :-
(1) q = w = 500 J, $\Delta U = 0$
(2) q = ΔU = 500 J, w = 0
(3) q = 0, w = 500 J, $\Delta U = 0$
(4) $\Delta U = 0$, q = w = – 500 J
CTD303

31. For which one of the following equations is $\Delta H^\circ_{reaction}$ equal to ΔH°_f for the product :-
(1) $N_2(g) + O_3(g) \rightarrow N_2O_3(g)$
(2) $CH_4(g) + 2Cl_2(g) \rightarrow CH_2Cl_2(l) + 2HCl(g)$
(3) $Xe(g) + 2F_2(g) \rightarrow XeF_4(g)$
(4) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$
CTD304

32. Internal energy of an ideal gas depends on:-
(1) Volume (2) Temperature
(3) Pressure (4) None of these
CTD305

33. The process in which no heat enters or leaves the system is terms as :-
(1) Isochoric (2) Isobaric
(3) Isothermal (4) Adiabatic
CTD306

34. The work done during the expansion of a gas from volume of 4 dm^3 to 6 dm^3 against a constant external pressure of 3 atm is :-
(1) +304 J (2) –304 J
(3) –6 J (4) –608 J
CTD307

35. Work done during reversible isothermal expansion of 1 mol of an ideal gas from 10 atm to 1 atm at 300K :-
(1) 4938.8 J (2) 4138.8 J
(3) 5744.1 J (4) 6257.2 J
CTD308

36. ($\Delta H - \Delta E$) for the formation of CO(g) from its elements at 298 K is :-
(1) –1238.78 J mol^{-1} (2) 1238.78 J mol^{-1}
(3) –2477.57 J mol^{-1} (4) 2477.57 J mol^{-1}
CTD309

37. Molar heat of vapourisation of a liquid is 6 kJ mol^{-1}. If the entropy change is 16 J mol^{-1} K^{-1}, the boiling point of the liquid is :-
(1) 375°C (2) 375 K
(3) 273 K (4) 102°C
CTD310

38. Heat of formation of $CO_2(g)$, $H_2O(\ell)$ and $CH_4(g)$ are –94, –68.4 and –17.9 kCal mol^{-1} respectively. the heat of combustion of $CH_4(g)$ is :-
(1) –212.9 kCal mol^{-1} (2) –136.8 kCal mol^{-1}
(3) –304.3 kCal mol^{-1} (4) –105.2 kCal mol^{-1}
CTD311

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39. Change in internal energy when 4 kJ work is done on the system and 1 kJ of heat is given by the system is :-
(1) +1 kJ (2) –5 kJ
(3) +5 kJ (4) +3 kJ

CTD312

40. In a chemical reaction ΔH is 150 kJ and ΔS is 100 J K^{-1} at 300 K then ΔG is :-
(1) 200 kJ (2) 333 kJ
(3) 300 kJ (4) 120 kJ

CTD313

41. Standard enthalpies of formation of O_3, CO_2, NH_3 and HI are 142.2, –393.2, –46.2 and +25.9 kJ mol^{-1} respectively. The order of their increasing stabilities will be :-
(1) O_3, CO_2, NH_3, HI
(2) CO_2, NH_3, HI, O_3
(3) O_3, HI, NH_3, CO_2
(4) NH_3, HI, CO_2, O_3

CTD314

42. The amount of heat released when 20 mL of 0.5 M NaOH is mixed with 100 mL of 0.1 M HCl is x kJ. The heat of neutralisation (in kJ mol^{-1}) is :-
(1) –100 x (2) –50 x
(3) +100 x (4) +50x

CTD315

43. What will be the heat of formation of methane, if the heat of combustion of carbon is '–x' kJ, heat of formation of water is '–y' kJ and heat of combustion of methane is '–z' kJ ?
(1) (– x – y + z) kJ
(2) (– z + x + 2y) kJ
(3) (– x – 2y – z) kJ
(4) (– x – 2y + z) kJ

CTD316

44. How much energy is absorbed by 10 mol of an ideal gas if it expands reversibly from an initial pressure of 8 atm. to 4 atm at a constant temperature of 27°C ?
(1) 1.728×10^4 J (2) 2×10^5 J
(3) 8×10^6 J (4) 1.728×10^5 J

CTD317

45. Which reaction occurs with the greatest increase in entropy ?
(1) $2H_2O(\ell) \rightarrow 2H_2(g) + O_2(g)$
(2) $2\,NO(g) \rightarrow N_2(g) + O_2(g)$
(3) $C(s) + O_2(g) \rightarrow CO_2(g)$
(4) $Br_2(g) + Cl_2(g) \rightarrow 2\,BrCl(g)$

CTD318

46. In which of the reactions at constant pressure sign of work done is –ve.
(1) $H_2O(s) \longrightarrow H_2O(g)$
(2) $H_2O(l) \longrightarrow H_2O(s)$
(3) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$
(4) All of these

CTD319

47. (i) $2B_{(s)} + \frac{3}{2}O_{2(g)} \longrightarrow B_2O_{3(s)}$
$\Delta H = -1273$ KJ mol^{-1}

(ii) $H_{2(g)} + \frac{1}{2}O_{2(g)} \longrightarrow H_2O_{(\ell)}$
$\Delta H = -286$ KJ mol^{-1}

(iii) $H_2O_{(\ell)} \longrightarrow H_2O_{(g)}$ $\Delta H = 44$ KJ mol^{-1}

(iv) $2B_{(s)} + 3H_{2(g)} \longrightarrow B_2H_{6(g)}$
$\Delta H = 36$ KJ mol^{-1}

From the above data, calculate the enthalpy change for the combustion of $B_2H_{6(g)}$.
$B_2H_{6(g)} + 3O_{2(g)} \longrightarrow B_2O_{3(s)} + 3H_2O_{(g)}$
(1) –1835 KJ mol^{-1} (2) 2035 KJ mol^{-1}
(3) –2035 KJ mol^{-1} (4) None of these

CTD320

48. The entropies of $H_2(g)$ and H(g) are 130.6 and 114.6 Jmol^{-1} K^{-1} respectively at 298K. Bond energy of H_2 (in kJ mol^{-1}) J is-
Given-
$H_2(g) \longrightarrow 2H(g)$; $\Delta G° = 406.62$ kJ mol^{-1}
(1) 364 kJ (2) 643 kJ
(3) 436 kJ (4) 346 kJ

CTD321

49. The latent heat of vaporisation of a liquid at 500 K and 1 atm pressure is 10 kcal/mol. What will be the change in internal energy (ΔU) during vaporisation of 3 moles of the liquid at the same temperature and pressure?
(1) 13 kcal (2) –13 kcal
(3) 27 kcal (4) –27 kcal

CTD322

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50. One gram sample of NH_4NO_3 is decomposed in a bomb calorimeter. The temperature of the calorimeter increase by 6.12 K. The heat capacity of the system is 1.23 kJ/g/K. What is the molar heat of decomposition for NH_4NO_3

(1) 7.53 kJ/mol
(2) 398.1 kJ/mol
(3) 16.1 kJ/mol
(4) 602 kJ/mol

CTD323

51. Consider the reaction at 300 K

$H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl_{(g)}$, $\Delta H° = -185$ kJ mol^{-1}

If 3 mole of H_2 completely react with 3 mol of Cl_2 to form HCl, $\Delta U°$ of the reaction will be :-

(1) Zero (2) –185 kJ
(3) –555 kJ (4) –200 kJ

CTD324

52. For which process will $\Delta H°$ and $\Delta G°$ be expected to be most similar :-

(1) $2Al(s) + Fe_2O_3(s) \rightarrow 2Fe(s) + Al_2O_3(s)$
(2) $2Na(s) + 2H_2O(I) \rightarrow 2NaOH(aq.) + H_2(g)$
(3) $2NO_2(g) \rightarrow N_2O_4(g)$
(4) $2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$

CTD325

53. Consider the following statements :

I. Change in state function between two states is a definite quantity and does not depend on path.
II. Intensive properties can't be algebraically added or subtracted.
III. Ratio of two extensive properties result into a parameter that depends on amount of substance.
IV. Molar heat capacity is an intensive property.

The correct order of true / false of the above statements is

(1) F T F T
(2) F F F T
(3) T T F T
(4) T T T F

CTD326

54. Consider the following statements :-

I. ΔG is negative for a physical process $H_2O_{(l)} \rightarrow H_2O_{(s)}$ at – 20°C.
II. Molar entropy of vapourisation of ice is negative.
III. In reversible adiabatic process, heat involved is zero and therefore entropy change will be zero.
IV. At absolute zero temperature, entropy of a pure perfectly crystalline substance is zero.

The correct order of true / false of the above statements is

(1) T T T T (2) T F T T
(3) F F F T (4) T F F T

CTD327

55. ($\Delta H - \Delta E$) is maximum at a given temperature in case of :-

(1) $PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$
(2) $CaCO_3(s) \rightarrow CaO_3(s) + CO_2(g)$
(3) $NH_4HS(s) \rightarrow NH_3(g) + H_2S(g)$
(4) $N_2(g) + O_2(g) \rightarrow 2NO(g)$

CTD328

56. Which has the highest entropy per mol of the substance :-

(1) H_2 at 25°C and 1 atm
(2) H_2 at STP
(3) H_2 at 100 K and 1 atm
(4) H_2 at 0 K and 1 atm

CTD329

57. One mol of NaCl(s) on melting absorbed 30.5 kJ of heat and its entropy is increased by 28.8 J K^{-1}. What is the melting point of sodium chloride :-

(1) 750 K (2) 1059 K
(3) 1332 K (4) 786 K

CTD330

58. Temperature of 1 mol of a gas is increased by 1° C at constant pressure. The magnitude of work done is :-

(1) R (2) 2 R
(3) $\frac{R}{2}$ (4) 3 R

CTD331

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59. Use the bond energies in the table to calculate ΔH for the reaction :

$$CH_2 = CH_2 + Cl_2 \longrightarrow \begin{matrix} CH_2 & — & CH_2 \\ | & & | \\ Cl & & Cl \end{matrix}$$

Bond	C-C	C=C	C-Cl	C-H	Cl-Cl
Bond energy (KJ/mole)	347	612	331	414	243

(1) $\Delta H = -684$ KJ
(2) $\Delta H = -154$ KJ
(3) $\Delta H = +89$ KJ
(4) $\Delta H = +177$ KJ

CTD332

60. When one mole of an ideal gas is compressed to half its initial volume and simultaneously heated to twice its initial temperature, the change in entropy (ΔS) is :-

(1) $C_V \ln 2$ (2) $C_P \ln 2$
(3) $R \ln 2$ (4) $(C_v - R) \ln 2$

CTD333

61. What are the signs of the entropy change (+ or –) in the following :-

I : A liquid crystallises into a solid
II : Temperature of a crystalline solid is raised from 0 K to 115 K
III : $2NaHCO_3(s) \rightarrow Na_2CO_3(s)+CO_2(g)+H_2O(g)$
IV : $H_2(g) \rightarrow 2H(g)$

	I	II	III	IV
(1)	–	+	+	+
(2)	–	–	–	+
(3)	–	–	–	+
(4)	+	–	–	–

CTD334

62. Consider following reactions :-

I : $N_2 + O_2 \rightarrow 2NO$ $\Delta H = x_1$
II : $2CO + O_2 \rightarrow 2CO_2$ $\Delta H = x_2$
III : $2H_2O + O_2 \rightarrow 2H_2O_2$ $\Delta H = x_3$
IV : $PCl_3 + Cl_2 \rightarrow PCl_5$ $\Delta H = x_4$

In which case(s), $\Delta H_f = \frac{\Delta H}{2}$

(1) I, II, III (2) IV
(3) II, III (4) I

CTD335

63. Calorific value of H_2 is – 143 kJ g^{-1}, Hence, ΔH_f° of H_2O is :-

(1) –143 kJ mol^{-1} (2) –286 kJ mol^{-1}
(3) +143 kJ mol^{-1} (4) +286 kJ mol^{-1}

CTD336

64. Heat of combustion of CH_4 at 25°C and 1 atm constant pressure is – 210 kcal mol^{-1}

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

Hence, heat of combustion at constant volume is :-

(1) –210 kCal mol^{-1}
(2) –198 kCal mol^{-1}
(3) –208.8 kCal mol^{-1}
(4) –211.2 kCal mol^{-1}

CTD337

65. In the following reaction, involving neutralisation of HF (a weak acid) with NaOH (a strong base).

$$HF(aq.) + NaOH(aq.) \rightarrow NaF(aq.) + H_2O(l)$$
$$\Delta H^\circ = -68.6 \text{ kJ mol}^{-1}$$

This values is much higher than the heat of neutralisation of strong acid with strong base (= – 57.3 kJ mol^{-1}). This is due to :

(1) In aqueous solution, HF behaves as a strong acid
(2) There is hydration of F^- ion in aqueous solution, and the process is exothermic
(3) Both (1) and (2)
(4) Can't be predicted

CTD338

66. In a closed container of 1.0 litre capacity 2 mole of CO and 1 mole of O_2 gases at 500 K and 70 atm react to form 2 mole of CO_2 gas. The pressure decreases to 40 atm at 500 K due to the following reaction :

$$2CO + O_2 \longrightarrow 2CO_2; \Delta H = -560 \text{ kJ}$$

Calculate ΔU for the reaction at 500 K. 1 litre-atm = 0.1 kJ. All the above gases show significant deviations from ideal gas behaviour.

(1) –557 kJ (2) –590 kJ
(3) +557 kJ (4) +590 J

CTD339

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67. The enthalpy change for which of the following processes represents the enthalpy of formation of AgCl :-

(1) $Ag^+(aq.) + Cl^-(aq.) \rightarrow AgCl(s)$

(2) $Ag(s) + AuCl(s) \rightarrow Au(s) + AgCl(s)$

(3) $Ag(s) + \frac{1}{2} Cl_2(g) \rightarrow AgCl(s)$

(4) $AgCl(s) \rightarrow Ag(s) + \frac{1}{2} Cl_2(s)$

CTD340

68. In the reactions

I : $H_2 \rightarrow 2H$, $\Delta H = 436$ kJ

II : $H_2 + \frac{1}{2} O_2 \rightarrow H_2O$, $\Delta H = -241.81$ kJ

III : $2H + \frac{1}{2} O_2 \rightarrow H_2O$, $\Delta H = ?$

H_2O can be formed either by II or III. Ratio of enthalpy change in III to II is :-

(1) 2.8 (2) 0.36

(3) 1.2 (4) 0.82

CTD341

69. 5 moles of a gas (assuming ideal) expand adiabatically and reversibly by doing work of 250 R and undergo the decrease in temperature of 20°C. The gas is :-

(1) He (2) CO

(3) NH_3 (4) CO_2

CTD342

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	4	3	3	2	2	4	2	2	2	1	1	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	3	2	2	3	1	3	1	1	2	3	2	1	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	2	4	4	3	2	2	1	4	4	3	1	4	1	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	3	3	3	4	3	1	3	2	3	1	2	1	2	4
Question	61	62	63	64	65	66	67	68	69						
Answer	1	4	2	3	2	1	3	1	2						

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Chemical Equilibrium

1. 48 g of SO_3, 12.8 g of SO_2 and 9.6 g of O_2 are present in one litre. The respective active masses will be

(1) 1.0, 0.5 and 0.3 (2) 0.6, 0.2 and 0.3
(3) 0.6, 0.4 and 0.2 (4) 1.0, 0.5 and 1.5

CCE142

2. If the forward rate constant of reversible reaction is 0.16 and backward rate constant is 4×10^4, then the equilibrium constant will be.

(1) 2.25×10^{-6} (2) 2.5×10^{-5}
(3) 4×10^{-6} (4) 4×10^{-5}

CCE143

3. If 0.5 mole H_2 is reacted with 0.5 mole I_2 in a 10 litre container at 444°C and At same temperature value of equilibrium constant K_c is 49, the ratio of [HI] and [I_2] will be :-

(1) 7 (2) $\frac{1}{7}$ (3) $\sqrt{\frac{1}{7}}$ (4) 49

CCE144

4. One mole of A reacts with one mole of B to form 0.1 mole of C and 0.1 mole of D. What will be the value of the equilibrium constant ?

$A + B \rightleftharpoons C + D$

(1) 0.1 (2) 100 (3) $\frac{1}{81}$ (4) 81

CCE145

5. For a reaction $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ at 721°C the value of equilibrium constant is 50. If 0.5 moles each of H_2 & I_2 is added to the system the value of equilibrium constant will be.

(1) 100 (2) 25 (3) 50 (4) 200

CCE146

6. In the reaction $A + 3B \rightleftharpoons 2C$, initially 10 mole each of A and B were taken. If equilibrium pressure is 8 atm and α of A = 20% then K_p will be :-

(1) $\frac{1}{4}$ (2) $\frac{1}{8}$ (3) $\frac{1}{16}$ (4) $\frac{1}{2}$

CCE147

7. In the reaction $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ at equilibrium the concentrations of N_2, O_2 and NO are 0.25, 0.05 and 1 M respectively. Calculate initial concentrations of N_2 and O_2.

(1) 0.25 M, 0.35 M (2) 0.05 M, 0.35 M
(3) 0.75 M, 0.55 M (4) 0.75 M, 0.45 M

CCE148

8. At the equilibrium of the reaction,
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$,
the observed molar mass of N_2O_4 is 77.70 g. The percentage dissociation of N_2O_4 is :-

(1) 28.4 (2) 46.7
(3) 22.4 (4) 18.4

CCE149

9. $A \rightleftharpoons 2B$, K_p; $C \rightleftharpoons D + E$, $K_{p'}$. If degrees of dissociation of A and C are same and $K_p = 2K_{p'}$, then the ratio of total pressure P/P'=?

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{1}{4}$ (4) 2

CCE150

10. When sulphur is heated at 800 K, the initial pressure of 1 atm decreases by 20% at equilibrium. K_p of the reaction $S_8(g) \rightleftharpoons 4S_2(g)$ is :-

(1) 0.2 (2) 0.512
(3) 1.51 (4) 2

CCE151

11. K_p value for
$C_2H_{4(g)} + H_{2(g)} \rightleftharpoons C_2H_{6(g)}$ is 5×10^{18} atm^{-1} and K_p for $C_2H_2 + H_{2(g)} \rightleftharpoons C_2H_{4(g)}$ is 5×10^{26} atm^{-1} at 50°C. Calculate K_p for the reaction
$2H_{2(g)} + C_2H_{2(g)} \rightleftharpoons C_2H_{6(g)}$

(1) 2.5×10^{45} (2) 21×10^8
(3) 1.5×10^{-45} (4) 2.5×10^{-45}

CCE152

12. For the reaction
$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$; $K_p = 0.09$.
The total pressure at equilibrium is :-

(1) 0.3 atm (2) 0.09 atm
(3) 0.6 atm (4) 0.36 atm

CCE153

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13. The conversion of ozone into oxygen is exothermic. Under what conditions is ozone the most stable ?

$$2O_3(g) \rightleftharpoons 3O_2(g)$$

(1) At low pressure and low temperature
(2) At high pressure and high temperature
(3) At high pressure and low temperature
(4) At low pressure and high temperature

CCE154

14. Consider the following gas-phase reaction :

$2A(g) + B(g) \rightleftharpoons C(g) + D(g)$

An equilibrium mixture of reactants and products is subjected to the following changes:-

(I) A decrease in volume
(II) An increase in temperature
(III) Addition of reactants
(IV) Addition of an inert gas at constant volume

Which of these changes affect the composition of the equilibrium mixture but leaves the value of K_c unchanged :

(1) I, II (2) I, II, III
(3) II, III, IV (4) I, III

CCE155

15. In the dissociation of $I_2(g)$ at 1000 K in a container of 1 litre :-

$I_2(g) \rightleftharpoons 2I^-(g)$; $K_C = 10^{10}$

Select the correct relation:-

(1) $[I_2] > [I^-]$ (2) $[I_2] < [I^-]$
(3) $[I_2] = [I^-]$ (4) $[I_2] = \frac{1}{2}[I^-]$

CCE156

16. Volume of the flask in which species are transferred is double of the earlier flask. In which of the following cases, equilibrium is affected?

I : $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
II : $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$
III : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
IV : $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$

(1) I, II (2) II, III
(3) I, III (4) III, IV

CCE157

17. In a reversible reaction $K_c > K_p$ and $\Delta H = +40$ kcal. The product will be obtained in less amount at :-

(1) High pressure, high temperature
(2) Low pressure, low temperature
(3) High pressure, low temperature
(4) Low pressure, high temperature

CCE158

18. For the dissociation reaction

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$,

the degree of dissociation (α) in terms of K_p and total equilibrium pressure P is

(1) $\alpha = \sqrt{\frac{4P + K_p}{K_P}}$

(2) $\alpha = \sqrt{\frac{K_p}{4P + K_P}}$

(3) $\alpha = \sqrt{\frac{K_p}{4P}}$

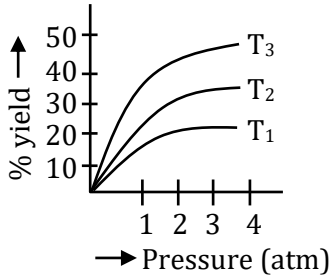
(4) None of these

CCE159

19. The preparation of $SO_3(g)$ by reaction

$$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$$

is an exothermic reaction. If the preparation follows the following temperature-pressure relationship for its % yield, then for temperatures T_1, T_2 & T_3. The correct option is



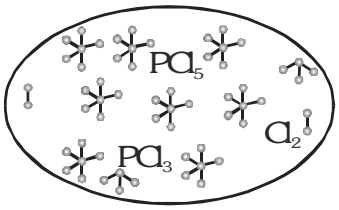
(1) $T_3 > T_2 > T_1$
(2) $T_1 > T_2 > T_3$
(3) $T_1 = T_2 = T_3$
(4) Nothing could be predicted about temperature through given information.

CCE160

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20. The equilibrium constant for :
$PCl_5 \rightleftharpoons PCl_3 + Cl_2$ is 0.5 at 500 K.
What is the state of vessel shown below ?



(1) The contents of the vessel are not in equilibrium, with an excess of PCl_5.
(2) The contents of vessel are in equilibrium.
(3) The contents of the vessel are not in equilibrium, with an excess of PCl_3 and Cl_2.
(4) Nothing can be predicted regarding the equilibrium just from the diagram given above.

CCE161

21. At 473 K, K_c for the reaction
$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ is 8.3×10^{-3}. What will be the value of K_c for the formation of PCl_5 at the same temperature ?
(1) 8.3×10^{3} (2) 120.48
(3) 8.3×10^{-3} (4) 240.8

CCE162

22. 18.4 g of N_2O_4 is taken in a 1 L closed vessel and heated till the equilibrium is reached.

$$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$$

At equilibrium it is found that 50% of N_2O_4 is dissociated. What will be the value of equilibrium constant ?
(1) 0.2 (2) 2 (3) 0.4 (4) 0.8

CCE163

23. 5 moles of SO_2 and 5 moles of O_2 react in a closed vessel. At equilibrium 60% of the SO_2 is consumed. The total number of gaseous moles (SO_2, O_2 and SO_3) in the vessel is :-
(1) 5.1 (2) 3.9 (3) 10.5 (4) 8.5

CCE164

24. For the reaction $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$, the value of K_c at 800°C is 0.1. What is the value of K_p at this temperature ?
(1) 0.5 (2) 0.01 (3) 0.05 (4) 0.1

CCE165

25. For the reaction a + b ⇌ c + d, initially concentrations of a and b are equal and at equilibrium the concentration of d is found to be twice of concentration of a. What will be the equilibrium constant for the reaction ?
(1) 2 (2) 9 (3) 4 (4) 3

CCE166

26. For a reaction
$NH_4COONH_{2(s)} \rightleftharpoons 2NH_{3(g)} + CO_{2(g)}$,
the equilibrium pressure is 3 atm. K_p for the reaction will be :-
(1) 27 (2) 4 (3) 3 (4) 9

CCE167

27. 0.6 moles of PCl_5, 0.3 mole of PCl_3 and 0.5 mole of Cl_2 are taken in a 1 L flask to obtain the following equilibrium;

$$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$$

If the equilibrium constant K_c for the reaction is 0.2. Predict the direction of the reaction.
(1) Forward direction
(2) Backward direction
(3) Direction of the reaction cannot be predicted
(4) Reaction does not move in any direction

CCE168

28. $N_2O_{4(g)} \rightleftharpoons 2NO_2$; $K_c = 5.7 \times 10^{-9}$ at 298 K
At equilibrium :-
(1) concentration of NO_2 is higher than that of N_2O_4
(2) concentration of N_2O_4 is higher than that of NO_2
(3) both N_2O_4 and NO_2 have same concentration
(4) concentration of N_2O_4 and NO_2 keeps on changing

CCE169

29. The following reaction is at equilibrium,

$$Fe^{3+}_{(aq)} + SCN^{-}_{(aq)} \rightleftharpoons [Fe(SCN)]^{2+}_{(aq)}$$

Yellow Colourless Deep red

$$K_c = \frac{[Fe(SCN)]^{2+}}{[Fe^{3+}]\,[SCN^-]}$$

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In the above reaction, colour intensity of red colour can be increased by :-

(1) addition of SCN^-

(2) addition of $H_2C_2O_4$ which reacts with Fe^{3+} ions

(3) addition of Hg^{2+} ions which react with SCN^- ions

(4) red colour intensity cannot be changed

CCE170

30. Formation of ClF_3 from Cl_2 and F_2 is an exothermic process. The equilibrium system can be represented as

$Cl_{2(g)} + 3F_{2(g)} \rightleftharpoons 2ClF_{3(g)}$; $\Delta H = -329$ kJ

Which of the following will increase quantity of ClF_3 in the equilibrium mixture ?

(1) Increase in temperature, decrease in pressure, addition of Cl_2

(2) Decrease in temperature and pressure, addition of ClF_3

(3) Increase in temperature and pressure, removal of Cl_2

(4) Decrease in temperature, increase in pressure, addition of F_2

CCE171

31. For the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

Initial moles are a b c

If α is the degree of dissociation and P is the total pressure, then the partial pressure of PCl_3 is :-

(1) $\frac{b+a\alpha}{a+b+c+a\alpha}$ P atm.

(2) $\frac{a+a\alpha}{a+b+c+a\alpha}$ P atm.

(3) $\frac{a(1-\alpha)}{a+b+c+a\alpha}$ P atm.

(4) $\frac{a+b+c+a\alpha}{b+a\alpha}$ P atm.

CCE172

32. $NaHCO_3$ decomposes as :

$2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + CO_2(g) + H_2O(g)$

The equilibrium pressure is 1.04 atm. The K_P for the reaction is :

(1) 0.2704 (2) 2.704

(3) 27.04 (4) 270.4

CCE173

33. The equilibrium constant at a certain temp. for the reaction $A_2 + B_2 \rightleftharpoons 2AB$ is 2. Calculate the degree of dissociation of either A_2 or B_2 :

(1) 0.2 (2) 0.5

(3) $\frac{1}{(1+\sqrt{2})}$ (4) $\frac{\sqrt{2}}{(1+\sqrt{2})}$

CCE174

34. The equilibrium constant for the esterification reaction of acetic acid and ethyl alcohol at 100°C is 4. What percentage of alcohol has been esterified ?

(1) 45% (2) 20%

(3) 54% (4) 66%

CCE175

35. In which of the following case the reaction is nearest to completion ?

(1) $K_{eq} = 10^6$ (2) $K_{eq} = 10^2$

(3) $K_{eq} = 10^{-6}$ (4) $K_{eq} = 10^{-12}$

CCE176

36. Reaction $N_2O_3 \rightleftharpoons NO + NO_2$ was studied by taking 2 mol L^{-1} of N_2O_3 initially. At equilibrium degree of dissociation of N_2O_3 was found to be 10%. What is the ratio of equilibrium concentration of $[N_2O_3]$, [NO] and $[NO_2]$:

(1) 19 : 1 : 1 (2) 9 : 1 : 1

(3) 1 : 1 : 9 (4) 18 : 1 : 1

CCE177

37. The equilibrium constant for reaction $H_2(g) + CO_2(g) \rightleftharpoons H_2O(g) + CO(g)$ is 1·8 at 1000°C. If one mole of H_2 and 1·0 mole of CO_2 are placed in a one litre flask. The final equilibrium concentration of CO at 1000°C will be $\left(\sqrt{1.8} = 1.34\right)$:-

(1) 0·573 M (2) 0·386 M

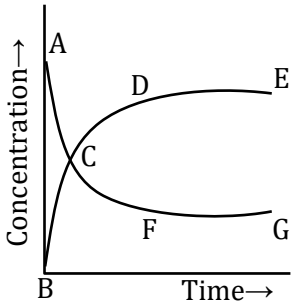
(3) 0·290 M (4) 0·688 M

CCE178

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38. $N_2O_4 \rightleftharpoons 2NO_2$, $K_c = 4$. This reversible reaction is studied graphically as shown in the given figure. Select the correct statements out of I, II and III



I. Reaction quotient has maximum value at point A

II. Reaction proceeds left to right at a point when $[N_2O_4] = [NO_2] = 0.1M$

III. $K_c = Q$ when point D or F is reached

(1) I, II (2) II, III

(3) I, III (4) I, II, III

CCE179

39. In the following equilibrium :–

I. $A + 2B \rightleftharpoons C$; K_1

II. $C + D \rightleftharpoons 3A$; K_2

III. $6B + D \rightleftharpoons 2C$; K_3 hence :-

(1) $3K_1 \times K_2 = K_3$

(2) $K_1^3 \times K_2^2 = K_3$

(3) $3K_1 \times K_2^2 = K_3$

(4) $K_1^3 \times K_2 = K_3$

CCE180

40. For the gas phase reaction,

$C_2H_{4(g)} + H_{2(g)} \rightleftharpoons C_2H_{6(g)}$;

$\Delta H = -136.8$ KJ mole^{-1}

carried out in a vessel, the concentration of C_2H_4 can be increased by

(a) increasing the temperature

(b) decreasing the pressure

(c) removing some H_2

(d) adding some C_2H_6

(1) (a), (b) (2) (b), (d)

(3) (a), (b), (c) (4) (a), (b), (c), (d)

CCE181

41. Consider the following reactions (in gaseous phase) in equilibrium with equilibrium concentrations 0.1 M of every species :-

(I) $2SO_2 + O_2 \rightleftharpoons 2SO_3$;

(II) $N_2 + 3H_2 \rightleftharpoons 2NH_3$;

(III) $N_2O_4 \rightleftharpoons 2NO_2$;

(IV) $4NO + 6H_2O \rightleftharpoons 4NH_3 + 5O_2$

Extent of reaction will be in order :

(1) I = II ≡ III = IV (2) I = II = III < IV

(3) III < I = IV < II (4) IV < III < I < II

CCE182

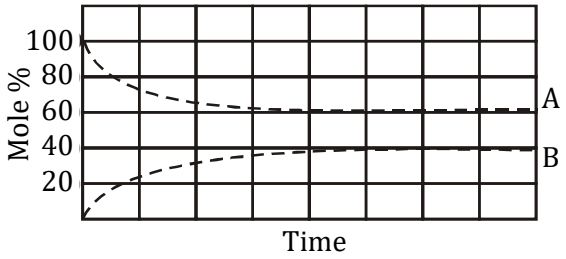
42. $2NOBr(g) \rightleftharpoons 2NO(g) + Br_2(g)$. If nitrosyl bromide (NOBr) is 40% dissociated at certain temp. and a total pressure of 0.30 atm. K_p for the reaction $2NO(g) + Br_2(g) \rightleftharpoons 2NOBr(g)$ is

(1) 45 (2) 25 (3) 0.022 (4) 0.25

CCE183

43. Consider the reaction : $A(g) \rightleftharpoons B(g)$.

One mole of A(g) is placed in a 1.0 L flask and the reaction is allowed to reach to equilibrium. A graph of the mole percent of A(g) and B(g) versus time is shown below :



The value of equilibrium constant is :

(1) 2 (2) 0.67 (3) 0.75 (4) 1.25

CCE184

44. The equilibrium constant for the reaction $H_2(g) + S(s) \rightleftharpoons H_2S(g)$; is 18.5 at 925 K and 9.25 at 1000 K respectively. The enthalpy of the reaction will be :

(1) –68000.05 J mol^{-1}

(2) –71080.57 J mol^{-1}

(3) –80071.75 J mol^{-1}

(4) 57080.75 J mol^{-1}

CCE185

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45. At a particular time, the analysis shows that composition of the reaction mixture is 3 mol lit^{-1} N_2, 2 mol lit^{-1} H_2 and 0.5 mol lit^{-1} NH_3 at 500 K. If equilibrium constant K_c for the reaction,

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ at 500 K is 0.061 lit^2 mol^{-2}, in which direction the net reaction will go:-

(1) Forward

(2) Backward

(3) Reaction is at equilibrium

(4) Direction of reaction can not be predicted

CCE186

46. For the dissociation reaction :

$$NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$$

if $K_P = 25\ atm^2$; then the equilibrium pressure of the system is :-

(1) 25 atm (2) 5 atm

(3) 10 atm (4) 15 atm

CCE187

47. The standard free energy change for a gaseous reaction at 27°C is X Kcal. If equilibrium constant for a reaction is 100 and R is 2 cal K^{-1} mol^{-1}. Then X is :-

(1) –2.7636 (2) –0.6736

(3) +2.2636 (4) None

CCE188

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**3**	**1**	**3**	**3**	**2**	**3**	**4**	**1**	**2**	**1**	**3**	**2**	**4**	**2**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**2**	**2**	**2**	**2**	**2**	**3**	**4**	**4**	**3**	**2**	**2**	**2**	**1**	**4**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**1**	**1**	**3**	**4**	**1**	**2**	**1**	**2**	**4**	**4**	**3**	**1**	**2**	**2**	**1**
Question	46	47													
Answer	**3**	**1**													

Ionic Equilibrium

1. The degree of hydrolysis of a salt of weak acid (HA) and weak base (BOH) in its 0.1 M solution is found to be 10%. If the molarity of the solution is 0.05 M, the percentage hydrolysis of the salt should be :-
(1) 5% (2) 10% (3) 20% (4) None
CIE276

2. The conjugate base of $(CH_3)_2\overset{+}{N}H_2$ is :-
(1) CH_3NH_2 (2) $(CH_3)_2N^+$
(3) $(CH_3)_2N$ (4) $(CH_3)_2NH$
CIE277

3. The two Bronsted bases in the reaction
$HC_2O_4^- + PO_4^{3-} \rightleftharpoons HPO_4^{2-} + C_2O_4^{2-}$ are
(1) $HC_2O_4^-$ and PO_4^{2-}
(2) HPO_4^{2-} and $C_2O_4^{2-}$
(3) PO_4^{3-} and $C_2O_4^{-2}$
(4) $HC_2O_4^-$ and HPO_4^{2-}
CIE278

4. Which of the following do not form conjugate acid-base pair in the right order :-
(1) NH_3, NH_2^- (2) OH^-, H_2O
(3) HCO_3^-, CO_3^{2-} (4) H_2S, HS^-
CIE279

5. Calculate pH of a solution whose 100 mL contains 0.2 g NaOH dissolved in it :-
(1) 10.699 (2) 11.699
(3) 12.699 (4) 13.699
CIE280

6. At 90°C, pH of an aq. solution of a strong electrolyte is 7. What is the nature of electrolyte ?
(1) neutral (2) basic
(3) acidic (4) none
CIE281

7. The pH of solution is increased from 3 to 6. Its H^+ ion concentration will be :-
(1) Reduced to half
(2) Doubled
(3) Reduced by 1000 times
(4) Increased by 1000 times
CIE282

8. Compute the pH of a solution at 25°C which is twice as alkaline as pure water.
(1) 6.7 (2) 6.4 (3) 7.3 (4) 7.6
CIE283

9. The pH of a solution containing NH_4OH and NH_4^+ is 9. If $[NH_4^+]$ = 0.1M and Ka of NH_4^+ is 5×10^{-10} then what is $[NH_4OH]$?
(1) 0.05 M (2) 20 M
(3) 0.5 M (4) 2 M
CIE284

10. The $[H_3O^+]$ in a solution having $[OH^-] = 5 \times 10^{-3}$ N is :-
(1) 2×10^{-3} N (2) 2×10^{-12} N
(3) 5×10^{-3} N (4) None
CIE285

11. Calculate $[H^+]$ of aq. solution containing equimolar CH_3COOH and CH_3COONa.
(pK_a CH_3COOH = 4.7) :-
(1) 1.2×10^9 (2) 7×10^{-5}
(3) 5×10^{-5} (4) 2×10^{-5}
CIE286

12. $FeCl_3$ solution is acidic due to hydrolysis of :-
(1) $FeCl_3$ (2) Cl^-
(3) Fe^{+3} (4) None of these
CIE287

13. pK_a of weak acid is 3.0. Effective range of a buffer of HA and A^- is about pH :-
(1) 3 to 4 (2) 3 to 6
(3) 2 to 3 (4) 2 to 4
CIE288

14. The pH of pure water at 80°C will be :-
(1) 7 (2) > 7
(3) < 7 (4) None of these
CIE289

15. Dissociation of NH_4OH is decreased in presence of :-
(1) CH_3COOH (2) KOH
(3) HNO_3 (4) None of these
CIE290

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16. The pH values 0.1 M solution of HCOONa (I), HCOOH (II), CH_3COONH_4 (III), NaOH (IV) HCl (V), will be in the order :-
(1) IV > III > I > IV > V
(2) IV > I > III > II > V
(3) II > III > I > IV > V
(4) V > II > III > I > IV

CIE291

17. Solution with reserve acidity and alkalinity are called :-
(1) Isohydric solution (2) True solution
(3) Normal solution (4) Buffer solution

CIE292

18. Which of the following expression is not applicable on the hydrolysis equilibrium :
$CN^- + H_2O \rightleftharpoons HCN + OH^-$
(1) $K_h = \frac{K_w}{K_{a(HCN)}}$ (2) $h = \sqrt{\frac{K_h}{C}}$
(3) $pH = \frac{1}{2} pK_{a(HCN)}$ (4) $[H^+] = \sqrt{\frac{K_w.K_a}{C}}$

CIE293

19. 150 mL of 0.0008 M ammonium sulphate is mixed with 50 mL of 0.04 M calcium nitrate. The ionic product of $CaSO_4$ will be ($K_{sp} = 2.4 \times 10^{-5}$ for $CaSO_4$) :-
(1) $< K_{sp}$ (2) $> K_{sp}$
(3) $\approx K_{sp}$ (4) None of these

CIE294

20. The solubility of $Ba_3(AsO_4)_2$ is 6.9×10^{-2} g/100 mL. What is the K_{sp}?
(formula weight = 690)
(1) 1.08×10^{-11} (2) 1.08×10^{-13}
(3) 1.0×10^{-15} (4) 6.0×10^{-13}

CIE295

21. Number of equivalent of HCl present in 100 mL of its solution whose pH is 4 :-
(1) 10^{-4} (2) 10^{-3} (3) 10^{-2} (4) 10^{-5}

CIE296

22. 0.01 M HA (aq.) is 2% ionized; $[OH^-]$ of solution is :-
(1) 2×10^{-4} (2) 10^{-8}
(3) 5×10^{-11} (4) 2×10^{-12}

CIE297

23. From separate solutions of four sodium salts NaW, NaX, NaY and NaZ has pH 7.0, 9.0, 10.0 and 11.0 respectively, when each solution was 0.1 M, the weakest acid is :-
(1) HW (2) HX (3) HY (4) HZ

CIE298

24. Silver ions are added to a solution with $[Br^-] = [Cl^-] = [CO_3^{2-}] = [AsO_4^{2-}] = 0.1$ M. Which compound will precipitate with lowest $[Ag^+]$?
(1) AgBr ($K_{sp} = 5 \times 10^{-13}$)
(2) AgCl ($K_{sp} = 1.8 \times 10^{-10}$)
(3) Ag_2CO_3 ($K_{sp} = 8.1 \times 10^{-12}$)
(4) Ag_3AsO_4 ($K_{sp} = 1 \times 10^{-22}$)

CIE299

25. Match the column I with column II and mark the appropriate choice.

Column I		Column II	
(A)	CH_3COONa	(i)	Depends on K_a and K_b
(B)	NH_4Cl	(ii)	Acidic, pH < 7
(C)	$NaNO_3$	(iii)	Alkaline, pH > 7
(D)	AgCN	(iv)	Neutral, pH = 7

(1) (A) → (i), (B) → (ii), (C) → (iii), (D) → (iv)
(2) (A) → (ii), (B) → (iii), (C) → (iv), (D) → (i)
(3) (A) → (iii), (B) → (ii), (C) → (iv), (D) → (i)
(4) (A) → (iv), (B) → (i), (C) → (iii), (D) → (ii)

CIE300

26. Solution of a mono basic acid has a pH = 5. If 1 mL of it is diluted to 1 litre, what will be the pH of the resulting solution ?
(1) 3.45 (2) 6.96
(3) 8.58 (4) 10.25

CIE301

27. The pH of 0.001 M $Ba(OH)_2$ solution will be :-
(1) 2 (2) 8.4
(3) 11.3 (4) 13

CIE302

28. NH_4CN is a salt of weak acid HCN ($K_a = 6.2 \times 10^{-10}$) and a weak base NH_4OH ($K_b = 1.8 \times 10^{-5}$). 1 molar solution of NH_4CN will be:-
(1) neutral (2) strongly acidic
(3) strongly basic (4) weakly basic

CIE303

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29. What is the pH of a solution obtained by mixing 10 mL of 0.1 M HCl and 40 mL of 0.2 M H_2SO_4?
(1) 0.74 (2) 7.4
(3) 4.68 (4) 0.468

CIE304

30. Match the column I with column II and mark the appropriate choice.

Column I		Column II	
(A)	$Fe(OH)_3$	(i)	$K_{sp} = s^2$
(B)	Ag_2CrO_4	(ii)	$K_{sp} = 27s^4$
(C)	CH_3COOAg	(iii)	$K_{sp} = 108s^5$
(D)	$Ca_3(PO_4)_2$	(iv)	$K_{sp} = 4s^3$

(1) (A) → (iii), (B) → (ii), (C) → (iv), (D) → (i)
(2) (A) → (ii), (B) → (iv), (C) → (i), (D) → (iii)
(3) (A) → (i), (B) → (iii), (C) → (ii), (D) → (iv)
(4) (A) → (iv), (B) → (i), (C) → (iii), (D) → (ii)

CIE305

31. Solubility product of radium sulphate is 4×10^{-11}. What will be the solubility of Ra^{2+} in 0.10 M Na_2SO_4 ?
(1) 4×10^{-10} M (2) 2×10^{-5} M
(3) 4×10^{-5} M (4) 2×10^{-10} M

CIE306

32. The solubility product of AgCl is 1.5625×10^{-10} at 25°C. Its solubility in g per litre will be:-
(1) 143.5 (2) 108
(3) 1.57×10^{-8} (4) 1.79×10^{-3}

CIE307

33. The solubility product of AgCl is 1.8×10^{-10}. Precipitation of AgCl will occur by mixing which of the following solutions are mixed in equal volume ?
(1) 10^{-8} M Ag^+ and 10^{-8} M Cl^- ions
(2) 10^{-3} M Ag^+ and 10^{-3} M Cl^- ions
(3) 10^{-6} M Ag^+ and 10^{-6} M Cl^- ions
(4) 10^{-10} M Ag^+ and 10^{-10} M Cl^- ions

CIE308

34. A solution containing Mn^{2+}, Fe^{2+}, Zn^{2+} and Hg^{2+} with a molar concentration of 10^{-3} M each is treated with 10^{-16} M sulphide ion solution. Which ions will precipitate first if K_{sp} of MnS, FeS, ZnS and HgS are 10^{-15}, 10^{-23}, 10^{-20} and 10^{-54} respectively ?
(1) FeS (2) MnS (3) HgS (4) ZnS

CIE309

35. What is the minimum concentration of SO_4^{2-} required to precipitate $BaSO_4$ in a solution containing 1×10^{-4} mole of Ba^{2+} ?
(K_{sp} for $BaSO_4 = 4 \times 10^{-10}$)
(1) 4×10^{-10} M (2) 2×10^{-10} M
(3) 4×10^{-6} M (4) 2×10^{-3} M

CIE310

36. Which of the following will produce a buffer solution when mixed in equal volume ?
(1) 0.1 mol dm^{-3} NH_4OH and 0.1 mol dm^{-3} HCl
(2) 0.05 mol dm^{-3} NH_4OH and 0.1 mol dm^{-3} HCl
(3) 0.1 mol dm^{-3} NH_4OH and 0.05 mol dm^{-3} HCl
(4) 0.1 mol dm^{-3} CH_3COONa and 0.1 mol dm^{-3} NaOH

CIE311

37. The molecular weight of compounds
(a) Na_2SO_4
(b) $Na_3PO_4 \cdot 12H_2O$
(c) $Ca_3(PO_4)_2$
respectively are X, Y and Z.
The correct set of equivalent weight will be :-
(1) $\frac{X}{2}, \frac{Y}{3}, \frac{Z}{6}$ (2) $\frac{X}{2}, Y, \frac{Z}{3}$
(3) $X, \frac{Y}{3}, \frac{Z}{3}$ (4) X, Y, Z

CIE312

38. pH of 0.01 M ammonium cyanide solution is 7.02. If 5 litres of water is added to this solution, the pH becomes :
(1) > 7.02
(2) < 7.02
(3) It is difficult to predict
(4) 7.02

CIE313

39. $pOH = 7 - 0.5\ pK_a + 0.5\ pK_b$, is true for which pair of cation and anion :
(1) $C_6H_5NH_3^+$, CH_3COO^-
(2) Na^+, CN^-
(3) Al^{+3}, Cl^-
(4) NH_4^+, NO_3^-

CIE314

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40. Expression to evaluate solubility for Hg_2Cl_2 is

(1) $\left[\frac{K_{sp}}{108}\right]^{1/5}$ (2) $\left[K_{sp}\right]^{1/2}$

(3) $\left[\frac{K_{sp}}{4}\right]^{1/3}$ (4) $\left[\frac{K_{sp}}{16}\right]^{1/4}$

CIE315

41. If the solubility of calcium phosphate (molecular weight M) in water at 25°C is w g per 100 mL, its solubility product at 25°C is approximately :

(1) $10^9\left(\frac{w}{M}\right)^5$ (2) $10^7\left(\frac{w}{M}\right)^5$

(3) $10^5\left(\frac{w}{M}\right)^5$ (4) $10^3\left(\frac{w}{M}\right)^5$

CIE316

42. HX is a weak acid ($K_a = 10^{-5}$). It forms a salt NaX (0.1 M) on reacting with caustic soda. The degree of hydrolysis of NaX is :

(1) 0.01% (2) 0.0001%

(3) 0.1% (4) 0.5%

CIE317

43. $CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$

In the above process, conjugate base of strong acid is :

(1) H_2O (2) CH_3COO^-

(3) CH_3COOH (4) H_3O^+

CIE318

44. The pH of a solution is 13. What is the number of H^+ ions in 1 mL of the solution ?

(1) 6.02×10^{23} (2) 6.02×10^{13}

(3) 6.02×10^{7} (4) 6.02×10^{10}

CIE319

45. Which of the following solution will have pH close to 1.0 ?

(1) 100 mL. of $\left(\frac{M}{10}\right)$ HCl + 100 mL of $\left(\frac{M}{10}\right)$ NaOH

(2) 55 mL. of $\left(\frac{M}{10}\right)$ HCl + 45 mL of $\left(\frac{M}{10}\right)$ NaOH

(3) 10 mL. of $\left(\frac{M}{10}\right)$ HCl + 90 mL of $\left(\frac{M}{10}\right)$ NaOH

(4) 75 mL. of $\left(\frac{M}{5}\right)$ HCl + 25 mL of $\left(\frac{M}{5}\right)$ NaOH

CIE320

46. Which of the following will act as a buffer :

(a) NaCl + NaOH

(b) Borax + boric acid

(c) $NaH_2PO_4 + Na_2HPO_4$

(d) $NH_4Cl + NH_4OH$

(1) b, c and d (2) b and c

(3) a, c and d (4) b and d

CIE321

47. Solubility product (K_{sp}) of salts of type MX, MX_2, M_3X at temperature T are 4×10^{-8}, 3.2×10^{-14} and 2.7×10^{-15} respectively. Solubility of the salts at temperature T are in the order :-

(1) $MX > MX_2 > M_3X$ (2) $MX_2 > M_3X > MX$

(3) $M_3X > MX_2 > MX$ (4) $MX > M_3X > MX_2$

CIE322

48. The amino acid glycine exists predominantly in the form $NH_3^+CH_2COO^-$ its conjugate base and conjugate acid will be respectively :

(1) $NH_2CH_2COO^-$, $NH_3^+CH_2COOH$

(2) NH_2CH_2COOH, $NH_2CH_2COO^-$

(3) $NH_3^+CH_2COOH$, $NH_2CH_2COO^-$

(4) None

CIE323

49. Consider the following :

A. Sodium acetate

B. Acetic acid + Sodium acetate (1 : 1)

C. Acetic acid

The pH of their equimolar aqueous solutions will be such that :

(1) C < B < A (2) B < A < C

(3) A < B < C (4) C < A < B

CIE324

50. Nicotinic acid ($K_a = 10^{-5}$) is represented by the formula HNiC. Calculate percent dissociation in a solution which contains 0.1 mol of nicotinic acid per 2 litre of solution :

(1) 1.4% (2) 1.6%

(3) 2.4% (4) 3.4%

CIE325

51. Calculate the pH at equivalence point when a solution of 0.1 M CH_3COOH is titrated with a solution of 0.1 M KOH.

(K_a of $CH_3COOH = 2 \times 10^{-5}$)

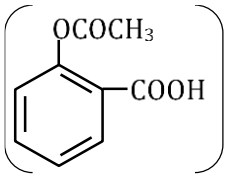
(1) 5.3 (2) 7.3 (3) 10.7 (4) 8.7

CIE326

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52. Acetyl salicylic acid called aspirin (shown in figure) is a pain killer with $pK_a = 4$. If two tablets each of 0.09 g mass containing aspirin are dissolved in 100 mL solution. Its pH will be



(1) 0.5 (2) 1.0 (3) 1.8 (4) 3

CIE327

53. Calculate the solubility of silver phosphate (Ag_3PO_4) in 0.1 M $AgNO_3$
(K_{sp} of $Ag_3PO_4 = 1.1 \times 10^{-16}$) :-
(1) 1×10^{-8} M (2) 4×10^{-15} M
(3) 1.1×10^{-15} M (4) 1.1×10^{-13} M

CIE328

54. A certain acidic indicator has pH = 5.48. 75% of the indicator exists in its ionised form. Calculate K_a of the indicator.
(1) 1×10^{-5} (2) 2×10^{-5}
(3) 1×10^{-3} (4) 4×10^{-5}

CIE329

55. Two sparingly soluble salts AX and BX_2 have their solubility product constants equal. Which of the following is incorrect ?
(1) Solubility of AX is less than solubility of BX_2
(2) If S_1 and S_2 are molar solubility of AX and BX_2 then $S_1 = 2(S_2)^{2/3}$
(3) If X is a conjugate base of a weak acid, addition of HNO_3 will increase solubility of both AX and BX_2
(4) Increasing the temperature, increases the solubility of both AX and BX_2

CIE330

56. What is the % dissociation of 0.01N NH_4OH in its buffer solution containing 0.01N NH_4Cl :-
(K_b of $NH_4OH = 2 \times 10^{-5}$)
(1) 0.2% (2) 2×10^{-3}%
(3) 2×10^{-5} (4) 5×10^{-3}%

CIE331

57. In which of the following combination, is buffer action expected ?
(a) $NH_4OH + NH_4Cl$
(b) HCl + NaCl
(c) NH_4OH + HCl in 2 : 1 mole ratio
Select the correct answer using the code given below :
(1) a & b (2) a & c
(3) a, b & c (4) None of these

CIE332

58. Find the pH of a solution obtained by mixing 25 mL, 0.5 M HCl solution, 10 mL, 0.5 M NaOH solution and 15 mL of water :-
(1) 1.82 (2) 0.48
(3) 0.82 (4) 12.1

CIE333

59. The pH of 0.5 M NaCN solution is (K_a of HCN = 5×10^{-10})
(1) 11.5 (2) 2.5
(3) 12.5 (4) 1.5

CIE334

60. What is the maximum concentration of equimolar solutions of $FeSO_4$ and Na_2S so that when mixed in equal volumes, there is no precipitation of FeS ?
(K_{sp} of FeS = 6.25×10^{-18})
(1) 5×10^{-9} M (2) 5×10^{-8} M
(3) 5×10^{-18} M (4) 5×10^{-19} M

CIE335

61. How many moles of HCN will be required to prepare one litre buffer solution of pH = 10.4 using 0.01 mole NaCN ?
(Given K_a(HCN) = 4×10^{-10})
(1) 2×10^{-2} mol (2) 1×10^{-3} mol
(3) 4×10^{-4} mol (4) 9×10^{-3} mol

CIE336

62. The pH of two equimolar weak acids are 3 and 5 respectively. Their relative strength is :-
(1) 3 : 5 (2) 5 : 3
(3) 100 : 1 (4) 1 : 100

CIE337

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63. 400 mL of 0.1 M KCN and 200 mL of 0.1 M HCl are mixed at 25°C. Determine the pH of the mixture. (pK_a for HCN = 6)

(1) 8 (2) 4

(3) 6 (4) 2

CIE338

64. How many gram of H_2SO_4 should be removed from 1 L of its aqueous solution of pH = 2 to increase its pH upto 3.

(1) 49 g (2) 9.8 g

(3) 0.49 g (4) 0.441 g

CIE339

65. 20 ml of 0.5 N HCl and 35 ml of 0.1 N NaOH are mixed. The resulting solution will :-

(1) be neutral

(2) be basic

(3) turn phenol phthalein solution pink

(4) turn methyl orange solution red

CIE340

66. The solubility of $Pb(OH)_2$ in water is 5×10^{-6} M. Calculate the solubility of $Pb(OH)_2$ in a buffer solution of pH = 8 at 25°C.

(1) 2.5×10^{-4} M (2) 5×10^{-16} M

(3) 5×10^{-4} M (4) 2.5×10^{-16} M

CIE341

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	3	2	3	2	3	3	1	2	4	3	4	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	3	1	2	4	3	4	1	3	2	3	4	4	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	4	2	3	3	3	1	4	1	3	2	1	1	3	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	4	1	1	1	4	4	4	1	2	1	2	3	1	1
Question	61	62	63	64	65	66									
Answer	2	3	3	4	4	3									

Electrochemistry

1. A 250 mL sample of a 0.2 M Cr^{3+} is electrolysed with a current of 96.5 amp; if the remaining concentration of Cr^{3+} ion is 0.1 M, the duration of electrolysis is : (Atomic wt. of Cr = 52) Assume volume of solution remains constant during electrolysis :-

(1) 75 sec (2) 150 sec
(3) 225 sec (4) 25 sec

CEL163

2. Electrolysis of hot aqueous solution of NaCl gives $NaClO_4$, i.e., sodium perchlorate,

$NaCl + 4H_2O \rightarrow NaClO_4 + 4H_2\uparrow$

How many faraday are required to obtain 1000 g of sodium perchlorate :-

(1) 66 (2) 40.3
(3) 18.3 (4) 31.6

CEL164

3. Which of the following equation is incorrect ?

(1) $\Lambda^{\circ}_{m(KCl)} - \Lambda^{\circ}_{m(NaCl)} = \Lambda^{\circ}_{m(KBr)} - \Lambda^{\circ}_{m(NaBr)}$

(2) $\Lambda^{\circ}_{m(KBr)} - \Lambda^{\circ}_{m(NaBr)} = \Lambda^{\circ}_{m(KI)} - \Lambda^{\circ}_{m(NaI)}$

(3) $\Lambda^{\circ}_{m(NaBr)} - \Lambda^{\circ}_{m(NaCl)} = \Lambda^{\circ}_{m(KBr)} - \Lambda^{\circ}_{m(NaI)}$

(4) $\Lambda^{\circ}_{m(NaCl)} - \Lambda^{\circ}_{m(NaI)} = \Lambda^{\circ}_{m(KCl)} - \Lambda^{\circ}_{m(NaI)}$

CEL165

4. Which of the following reaction is preferred at anode during the electrolysis of H_2SO_4 at higher concentration ?

(1) $2H_2O_{(\ell)} \longrightarrow O_{2(g)} + 4H^+ + 4e^-$

(2) $2SO^{-2}{}_{4(aq)} \longrightarrow S_2O_8^{-2}{}_{(aq)} + 2e^-$

(3) Both (1) and (2)
(4) None of the above

CEL166

5. Efficiency of the following cell is 84%.

$A(s) + B^{2+}(aq) \rightleftharpoons A^{2+}(aq) + B(s)$;

$\Delta H^\circ = -285$ kJ

Then the standard electrode potential of the cell will be :-

(1) 1.20 V (2) 2.40 V
(3) 1.10 V (4) 1.24 V

CEL167

6. The pressure of hydrogen gas is increased from 1 atm to 100 atm. Keeping the H^+ (1M) constant, the voltage of the hydrogen half-cell at 25ºC will be :-

(1) 0.059 V (2) 0.59 V
(3) 0.0259 V (4) – 0.0591 V

CEL168

7. For the cell reaction,

$Mg(s) + 2Ag^+(aq) \rightleftharpoons Mg^{2+}(aq) + 2Ag(s)$, E°_{cell} is 3.17 V at 298 K. The value of E_{cell}, ΔG° and Q at Ag^+ and Mg^{2+} concentrations of 0.001 M and 0.02 M respectively are :-

(1) 3.04 V, – 605.8 kJ mol^{-1}, 20000
(2) 3.04 V, 611.8 kJ mol^{-1}, 20000
(3) 3.13 V, – 604 kJ mol^{-1}, 20
(4) 3.04 V, – 611.8 kJ, 20000

CEL169

8. Consider the following four electrodes,

P = Cu^{2+} (0.0001 M) / Cu(s)
Q = Cu^{2+} (0.1 M) / Cu(s)
R = Cu^{2+} (0.01 M) / Cu(s)
S = Cu^{2+} (0.001 M) / Cu(s)

If the standard reduction potential of Cu^{2+} / Cu is 0.34 V, the reduction potential in volt of the above electrodes follow the order :-

(1) Q > R > S > P (2) P > S > R > Q
(3) R > S > Q > P (4) P > Q > R > S

CEL170

9. Which is correct about fuel cell :-

(1) Cell continuously run as long as fuels are supplied
(2) These are more efficient and free from pollution
(3) These are used to provide electrical power and drinking water to astronauts in space programme
(4) All of these

CEL171

10. The density of Cu is 8.94 g cm^{-3}. The quantity of electricity needed to plate an area 10 cm × 10 cm to a thickness of 10^{-2} cm using $CuSO_4$ solution would be.

(1) 13586 C (2) 27172 C
(3) 40758 C (4) 20348 C

CEL172

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11. Salts of A (at wt. 7), B (at wt. 27) and C(at wt. 48) were electrolysed under identical conditions using the same quantity of electricity. It was found that when 2.1 g of A was deposited, the weights of B and C deposited were 2.7 g and 7.2 g. The valencies of A, B and C are respectively :-

(1) 3, 1 and 2 (2) 1, 3 and 2

(3) 3, 1 and 3 (4) 2, 3 and 2

CEL173

12. The electrical resistance of a column of 0.05 mole/litre NaOH solution of diameter 1 cm and length 50 cm is 5.55×10^3 ohm. Calculate its molar conductivity :-

(1) 229.6 $Scm^2\ mol^{-1}$

(2) 119.5 $Scm^2\ mol^{-1}$

(3) 190.2 $Scm^2\ mol^{-1}$

(4) None of these

CEL174

13. A cell in which the following reaction takes place :-

$Mg_{(s)} + 2Ag^+ (0.0001\ M) \rightarrow Mg^{+2}(0.120\ M) + 2Ag(s)$

if $E^\circ_{cell} = 3.17$ volt then calculate E_{cell}

(1) 2.0 volt (2) 2.96 volt

(3) 3.50 volt (4) 3.17 volt

CEL175

14. An electrochemical cell is set up as follows' $Pt(H_2\ 1atm)/0.1\ M\ HCl\ ||\ 0.1\ M$ Acetic Acid/$(H_2\ 1atm)\ Pt$.

E.M.F. of the cell will not be zero because :

(1) The pH of 0.1 M HCl and 0.1 M acetic acid is not the same

(2) Acids used in the two compartments are different

(3) EMF of a cell depends on the molarities of acids used

(4) The temperature is constant

CEL176

15. Standard electrode potential of some half cell reactions are given below -

(A) $Sn^{+4} + 2e^- \longrightarrow Sn^{2+}$, $E^\circ = +0.15$ V

(B) $2Hg^{+2} + 2e^- \longrightarrow Hg_2^{+2}$, $E^\circ = +0.92$ V

(C) $PbO_2 + 4H^+ + 2e^- \longrightarrow Pb^{+2} + 2H_2O$, $E^\circ = +1.45$ V

for these which statement is correct –

(1) Pb^{+2} is more powerful reducing agent than Sn^{+2}

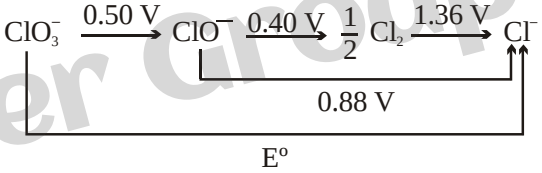
(2) Sn^{+4} is more powerful oxidising agent than Pb^{+4}

(3) Sn^{+2} is more powerful reducing agent than Hg_2^{+2}

(4) Hg^{+2} is more powerful oxidising agent than Pb^{+4}

CEL177

16. The E° in the given diagram is :-



(1) 0.5 (2) 0.63 (3) 0.7 (4) 0.8

CEL178

17. If same quantity of electricity is passed through CuCl and $CuSO_4$, the ratio of the weights of Cu deposited from $CuSO_4$ and CuCl is :-

(1) 2 : 1 (2) 1 : 2 (3) 1 : 1 (4) 4 : 1

CEL179

18. The $k = 4.95 \times 10^{-5}$ S cm^{-1} for a 0.00099 M solution. Calculate the degree of dissociation of acetic acid, if Λ_M^∞ for acetic acid is 400 S $cm^2\ mol^{-1}$:-

(1) 7 (2) 0.125 (3) 8 (4) 10

CEL180

19. Value of Λ_m^∞ for $SrCl_2$ in water at 25º from the following data :

Conc. (mol/L)	0.25	1
$\Lambda_m\ (\Omega^{-1}\ cm^2\ mol^{-1})$	260	250

(1) 270 (2) 260

(3) 250 (4) 255

CEL181

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20. The standard reduction potentials E° of the following reactions are

	E° (Volt)
(i) $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	1.51
(ii) $Sn^{4+} + 2e^- \rightarrow Sn^{2+}$	0.15
(iii) $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	1.33
(iv) $Ce^{4+} + e^- \rightarrow Ce^{3+}$	1.61

The oxidising power of the various species decreases in the order :-

(1) $Ce^{4+} > Cr_2O_7^{2-} > Sn^{4+} > MnO_4^-$

(2) $Ce^{4+} > MnO_4^- > Cr_2O_7^{2-} > Sn^{4+}$

(3) $Cr_2O_7^{2-} > Sn^{4+} > Ce^{4+} > MnO_4^-$

(4) $MnO_4^- > Ce^{4+} > Sn^{4+} > Cr_2O_7^{2-}$

CEL182

21. The reduction potential of a half-cell consisting of a Pt electrode immersed in 1.5 M Fe^{2+} and 0.015 M Fe^{3+} solution at 25° C is $\left(E^{\circ}_{Fe^{3+}/Fe^{2+}} = 0.770\ V\right)$

(1) 0.652 V (2) 0.88 V

(3) 0.710 V (4) 0.850 V

CEL183

22. The number of Faradays required to produce one mole of water from a hydrogen-oxygen fuel cell containing aqueous alkali as electrolyte is

(1) 1 (2) 3 (3) 2 (4) 4

CEL184

23. 4 mole of electrons were transferred from anode to cathode in an experiment on electrolysis of water. The total volume of the two dry gases (at STP) produced will be approximately (in L) :-

(1) 22.4 (2) 44.8 (3) 67.2 (4) 89.4

CEL185

24. The ionization constant of a weak electrolyte is 25×10^{-6} while the equivalent conductance of its 0.01 M solution is 19.6 S $cm^2\ eq^{-1}$. The equivalent conductance of the electrolyte at infinite dilution (in S $cm^2\ eq^{-1}$) will be :-

(1) 250 (2) 196 (3) 392 (4) 384

CEL186

25. Three moles of electrons are passed through three solutions in succession containing $AgNO_3$, $CuSO_4$ and $AuCl_3$, respectively. The molar ratio of amounts of cations reduced at cathode will be :-

(1) 1 : 2 : 3 (2) 2 : 1 : 3

(3) 3 : 2 : 1 (4) 6 : 3 : 2

CEL187

26. Given,

$Co^{3+} + e^- \longrightarrow Co^{2+}$; E° = +1.81 V

$Pb^{4+} + 2e^- \longrightarrow Pb^{2+}$; E° = +1.67 V

$Ce^{4+} + e^- \longrightarrow Ce^{3+}$; E° = +1.61 V

$Bi^{3+} + 3e^- \longrightarrow Bi$; E° = +0.20 V

Oxidising power of the species will increases in the order

(1) $Ce^{4+} < Pb^{4+} < Bi^{3+} < Co^{3+}$

(2) $Bi^{3+} < Ce^{4+} < Pb^{4+} < Co^{3+}$

(3) $Co^{3+} < Ce^{4+} < Bi^{3+} < Pb^{4+}$

(4) $Co^{3+} < Pb^{4+} < Ce^{4+} < Bi^{3+}$

CEL188

27. Calculate the standard cell potential (in V) of the cell in which following reaction takes place

$Fe^{2+}(aq) + Ag^+(aq) \longrightarrow Fe^{3+}(aq) + Ag(s)$

Given that, $E^0_{Ag^+/Ag} = xV$

$E^0_{Fe^{2+}/Fe} = yV$

$E^0_{Fe^{3+}/Fe} = zV$

(1) x + 2y–3z (2) x–y

(3) x+y–z (4) x–z

CEL189

28. Given, that

$E^{\Theta}_{S_2O_8^{2-}/SO_4^{2-}} = 2.05$; $E^{\Theta}_{Br_2/Br^{\Theta}} = +1.09V$, ;

$E^{\Theta}_{Au^{3+}/Au} = +1.4V$

The strongest oxidising agent is

(1) Au^{3+} (2) O_2 (3) $S_2O_8^{2-}$ (3) Br_2

CEL190

29. Consider the following reduction processes :

$Zn^{2+} + 2e^- \longrightarrow Zn(s)$; E° = –0.76 V

$Ca^{2+} + 2e^- \longrightarrow Ca(s)$; E° = –2.87 V

$Mg^{2+} + 2e^- \longrightarrow Mg(s)$; E° = –2.36 V

$Ni^{2+} + 2e^- \longrightarrow Ni(s)$; E° = –0.25 V

The reducing power of the metals increases in the order

(1) Zn < Mg < Ni < Ca

(2) Ni < Zn < Mg < Ca

(3) Ca < Zn < Mg < Ni

(4) Ca < Mg < Zn < Ni

CEL191

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30. The anodic half-cell of lead-acid battery is recharged using electricity of 0.05 Faraday. The amount of $PbSO_4$ electrolysed in g during the process is (Molar mass of $PbSO_4$ = 303g mol^{-1})

(1) 11.4 (2) 7.6 (3) 15.2 (4) 22.8

CEL192

31. For the following electrochemical cell at 298K

$Pt(s)|H_2(g, 1bar)|H^+(aq, 1M)$

$|| M^{4+}(aq), M^{2+}(aq)|Pt(s)$

$E_{cell} = 0.092V$ when $\frac{[M^{2+}(aq)]}{[M^{4+}(aq)]} = 10^x$

Given : $E^0_{M^{4+}/M^{2+}} = 0.151V; 2.303\frac{RT}{F} = 0.059V$

The value of x is

(1) –2 (2) –1 (3) 1 (4) 2

CEL193

32. Two Faraday of electricity is passed through a solution of $CuSO_4$. The mass copper deposited at the cathode is (at. mass of Cu = 63.5 u)

(1) 0 g (2) 63.5 g
(3) 2 g (4) 127 g

CEL194

33. Saturated solution of KNO_3 is used to make 'salt-bridge' because

(1) velocity of K^+ is greater than that of NO_3^-
(2) velocity of NO_3^- is greater than that of K^+
(3) velocities of both K^+ and NO_3^- are nearly the same
(4) KNO_3 is highly soluble in water

CEL195

34. The standard reduction potential values of three metallic cations, X, Y, Z are 0.52, –3.03 and –1.18 V respectively. The order of reducing power of the corresponding metals is.

(1) Y > Z > X (2) X > Y > Z
(3) Z > Y > X (4) Z > X > Y

CEL196

35. The standard reduction potentials E°, for the half reactions are as

$Zn = Zn^{2+} + 2e^-$, E° = +0.76 V
$Fe = Fe^{2+} + 2e^-$, E° = 0.41 V

The emf for the cell reaction,

$Fe^{2+} + Zn \longrightarrow Zn^{2+} + Fe$ is

(1) –0.35 V (2) +0.35 V
(3) +1.17 V (4) –1.17 V

CEL197

36. The reaction,

$\frac{1}{2}H_2(g) + AgCl(s) \rightleftharpoons H^+(aq) + Cl^-(aq) + Ag(s)$

occurs in the galvanic cell with representation:

(1) $Ag|AgCl(s)|KCl(soln)|AgNO_3|Ag$
(2) $Pt|H_2(g)|HCl(soln)|AgNO_3(soln)|Ag$
(3) $Pt|H_2(g)HCl(soln)||AgCl(s)|Ag$
(4) $Pt|H_2(g)|KCl(soln)|AgCl(s)|Ag$

CEL198

37. Faraday's laws of electrolysis are related to the

(1) atomic number of the cation
(2) atomic number of the anion
(3) equivalent weight of the electrolyte
(4) speed of the cation

CEL199

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**1**	**4**	**2**	**4**	**4**	**4**	**1**	**4**	**2**	**2**	**1**	**2**	**1**	**3**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**2**	**2**	**2**	**1**	**2**	**1**	**3**	**3**	**3**	**4**	**2**	**1**	**3**	**2**	**2**
Question	31	32	33	34	35	36	37								
Answer	**4**	**2**	**3**	**1**	**2**	**3**	**3**								

Solutions

1. How many milli moles of sucrose should be dissolved in 500 g of water so as to get a solution which has a difference of 103.57°C between boiling point and freezing point of solution :-
(K_f = 1.86 K kg mol^{-1}, K_b = 0.52 K kg mol^{-1})
(1) 500 milli moles (2) 900 milli moles
(3) 750 milli moles (4) 650 milli moles
CSO194

2. A solution of x moles of sucrose in 100 g of water freezes at – 0.2°C. Calculate the amount of ice separate out when it is cooled upto –0.25°C :-
(1) 18 g (2) 20 g (3) 25 g (4) 23 g
CSO195

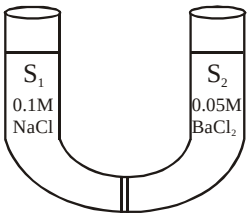
3. Which of the following has been arranged in order of decreasing freezing point :-
(1) 0.05 M KNO_3 > 0.04 M $BaCl_2$ > 0.140 M sucrose > 0.075 M $CuSO_4$
(2) 0.04 M $BaCl_2$ > 0.140 M sucrose > 0.075 M $CuSO_4$ > 0.05 M KNO_3
(3) 0.075 M $CuSO_4$ > 0.140 M sucrose > 0.04 M $BaCl_2$ > 0.05 M KNO_3
(4) 0.075 M $CuSO_4$ > 0.05 M KNO_3 > 0.140 M sucrose > 0.04 M $BaCl_2$
CSO196

4. The Van't Hoff factor for $BaCl_2$ at 0.01M concentration is 1.98. The percentage dissociation of $BaCl_2$ at this concentration is –
(1) 49 (2) 69 (3) 89 (4) 98
CSO197

5. 6.8 g H_2O_2 is dissolved in 224 mL solution. This solution will be labelled as :-
(1) 0.224 V (2) 20 V
(3) 5 V (4) 10 V
CSO198

6. The solubility of a gas in water at 300 K under a pressure of 100 atmospheres is 4 × 10^{-3} kg L^{-1}. Therefore, the mass of the gas in kg dissolved in 250 mL of water under a pressure of 250 atm and 300 K is :-
(1) 2.5 × 10^{-3} (2) 2.0 × 10^{-3}
(3) 1.25 × 10^{-3} (4) 5.0 × 10^{-3}
CSO199

7. Two solutions S_1 and S_2 containing 0.1 M NaCl (aq.) and 0.05 M $BaCl_2$ (aq.) are separated by semiperameable membrane.
Which among the following statement(s) is/are correct? (Assume complete dissociation of both the electrolytes)



(1) S_1 and S_2 are isotonic
(2) S_1 is hypertonic while S_2 is hypotonic
(3) S_1 is hypotonic while S_2 is hypertonic
(4) All the above
CSO200

8. The molarity of a solution made by mixing 50 mL of conc. H_2SO_4 (36 N) with 50 mL of water, is :
(1) 36 M (2) 18 M (3) 9 M (4) 6M
CSO201

9. An aqueous solution of urea containing 18 g urea in 1500 cm^3 of solution has a density of 1.052 g/cm^3. If the molecular mass of urea is 60, then the molality of solution is :-
(1) 0.2 (2) 0.192 (3) 0.064 (4) 1.2
CSO202

10. Which solution will have the highest boiling point :-
(1)1% w/v glucose in water
(2) 1% w/v sucrose in water
(3) 1% w/v NaCl in water
(4) 1% w/v $CaCl_2$ in water
CSO203

11. At 40°C, the vapour pressure (in torr) of methyl alcohol (A) and ethyl alcohol (B) solution is represented by :P = 120 X_A + 138; where X_A is mole fraction of methyl alcohol. The value of lim

$\lim_{x_A \to 0}\left(\frac{P_B^\circ}{X_B}\right)$ and $\lim_{x_B \to 0}\left(\frac{P_A^\circ}{X_A}\right)$ are :

(1) 138, 258 (2) 258,138
(3) 120, 138 (4) 138, 125
CSO204

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12. The vapour pressure of a solution of a non-volatile non-electrolyte (A) in a solvent (B) is 95 % of the vapour pressure of the solvent at the same temperature. If $M_B = 0.3\ M_A$, where M_B and M_A are molecular weight of B and A respectively, the weight ratio of the solvent and solute are :-

(1) –0.17 (2) 5.7 (3) 0.2 (4) 4.0

CSO205

13. When 20 g of naphthoic acid ($C_{11}H_8O_2$) is dissolved in 50g of benenze (K_f = 1.72 K kg mol^{-1}), then freezing point depression of 2K is observed. The van't Haff factor (i) is :

(1) 0.5 (2) 1.0 (3) 2.0 (4) 3.0

CSO206

14. If two substances A and B have $P_A^0 : P_B^0 = 1 : 2$ and have mole fraction in solution 1 : 2 then mole fraction of A in vapours

(1) 0.33 (2) 0.25 (3) 0.52 (4) 0.2

CSO207

15. The mass of solute 'A' (mol mass = 40 g mol^{-1}) that should be added to 180 g of pure water in order to lower its vapour pressure to $4/5^{th}$ of its original value :-

(1) 187.5 g (2) 90 g
(3) 150 g (4) 100 g

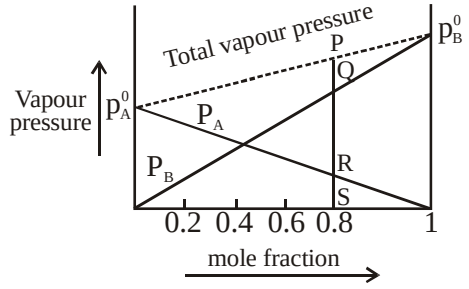
CSO208

16. 150mL of C_2H_5OH (density = 0.78 g/mL) is diluted to one litre by adding water. Molality of the solution is –

(1) 1.54 (2) 11.7 (3) 2.99 (4) 29.9

CSO209

17. Consider the following vapour pressure-composition graph, SP is equal to :-



(1) PQ + RS (2) QR + RS
(3) SR + SQ (4) PQ + QR

CSO210

18. pH of a 0.1 M monobasic acid is measured to be 2. Its osmotic pressure at a given temperature T K is :-

(1) 0.1 RT (2) 0.11 RT
(3) 1.1 RT (4) 0.01 RT

CSO211

19. A solution containing 0.1 g of a non-voltatile organic substance P(molecular mass 100) in 100g of benzene raises the boiling point of benzene by 0.2°C, while a solution containing 0.1 g of another non-volatile substance Q in the same amount of benzene raises the boiling point of benzene by 0.4°C. What is the ratio of molecular masses at P and Q ?

(1) 1 : 2 (2) 2 : 1 (3) 1 : 4 (4) 4 : 1

CSO212

20. What is the osmotic pressure of the solution obtained by mixing 300 cm^3 of 0.6% (mass-volume) solution of urea with 300 cm^3 3.42% solution of sucrose at 20°C ?

(1) 5 atm (2) 5.2 atm
(3) 2.4 atm (4) 4.5 atm

CSO213

21. Two solutions of a substance (non-electrolyte) are mixed in the following manner 480 mL of 1.5M of first solution with 520 mL of 1.2 M of second solution. The molarity of final solution is:-

(1) 1.20 M (2) 1.50 M
(3) 1.344 M (4) 2.70 M

CSO214

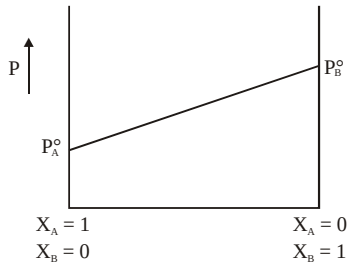
22. Equal amounts (mass) of a solute are dissolved in equal amounts of the solvents A and B. The relative lowering of vapour pressure for the solution A has twice that for the solution B. If M_A and M_B are the molar masses of solvent A and B respectively, then for dilute solution of A and B with solute :-

(1) $M_A = M_B$ (2) $M_A = \frac{M_B}{2}$
(3) $M_A = 4M_B$ (4) $M_A = 2M_B$

CSO215

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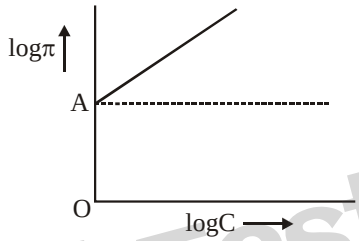
23. The following is a graph plotted between the vapour pressures of two volatile liquids against their respective mole fractions :-



(1) When $X_A = 1$ then $P = P^\circ_A$
(2) When $X_B = 1$ then $P < P^\circ_A$
(3) When $X_A = 1$ then $P < P^\circ_B$
(4) Both (1) and (3)

CSO216

24. Variation of $\log\pi$ (atm) v/s logC (C = concentration of solute in g/L) is given



At 300K, for a solute of molar mass 300 g/mol, OA = log ($x \times 10^{-2}$). What is the value of x ?

(1) 8.21 (2) 0.0821
(3) 82.1 (4) 0.821

CSO217

25. A 0.025 M solution of monobasic acid had a freezing point of –0.06°C. Calculate K_a for the acid. K_f for H_2O=1.86°C molality^{-1}. Assume molality equal to molarity.

(1) 1.96×10^{-3} (2) 3.96×10^{-3}
(3) 2.96×10^{-3} (4) 5.8×10^{-3}

CSO218

26. What would be the molality of 20% (mass/mass) aqueous solution of KI? (Molar mass of KI = 166 g mol^{-1})

(1) 1.48 (2) 1.51
(3) 1.35 (4) 1.08

CSO219

27. The vapour pressure of pure liquids A and B are 400 and 600 mmHg, respectively at 298K. On mixing the two liquids, the sum of their initial volumes is equal to the volume of the final mixture. The mole fraction of liquid B is 0.5 in the mixture. The vapour pressure of the final solution, the mole fractions of components A and B in vapour phase, respectively are

(1) 450 mmHg, 0.4, 0.6
(2) 500 mmHg, 0.5, 0.5
(3) 450 mmHg, 0.5, 0.5
(4) 500 mmHg 0.4, 0.6

CSO220

28. Liquids A and B form an ideal solution in the entire composition range. At 350K, the vapour pressures of pure A and pure B are 7×10^3 Pa and 12×10^3 Pa, respectively. The composition of the vapour in equilibrium with a solution containing 40 mole percent of A at this temperature is

(1) $x_A = 0.76$; $x_B = 0.24$
(2) $x_A = 0.28$; $x_B = 0.72$
(3) $x_A = 0.4$; $x_B = 0.6$
(4) $x_A = 0.37$; $x_B = 0.63$

CSO221

29. The Henry's law constant for the solubility of N_2 gas in water at 298 K is 1.0×10^5 atm. The mole fraction of N_2 in air is 0.8. The number of moles of N_2 from air dissolved in 10 moles of water of 298 K and 5 atm pressure is

(1) 4.0×10^{-4} (2) 4.0×10^{-5}
(3) 5.0×10^{-4} (4) 4.0×10^{-6}

CSO222

30. Liquids A and B ideal solution for all compositions of A and B at 25°C. Two such solutions with 0.25 and 0.50 mole fractions of A have the total vapour pressure of 0.3 and 0.4 bar, respectively. What is the vapour pressure of pure liquid B in bar ?

(1) 0.2 (2) 0.4 (3) 0.6 (4) 0.5

CSO223

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31. A solutions is prepared by dissolving 0.6 g of urea (molar mass = 60g mol^{-1}) and 1.8 g of glucose (molar mass = 180g mol^{-1}) in 100 mL of water at 27°C. The osmotic pressure of the solution is (R = 0.08206L atm K^{-1} mol^{-1})

(1) 8.2 atm (2) 2.46 atm

(3) 4.92 atm (4) 1.64 atm

CSO224

32. 1g of a non-volatile, non-electrolyte solute is dissolved in 100 g of two different solvents A and B, whose ebullisocopic constants are in the ratio of 1 : 5. The ratio of the elevation in their boiling points, $\frac{\Delta T_b(A)}{\Delta T_b(B)}$ is

(1) 5 : 1 (2) 10 : 1

(3) 1 : 5 (4) 1 : 0.2

CSO225

33. At room temperature, a dilute solution of urea is prepared by dissolving 0.60 g of urea in 360 g of water. if the vapour pressure of pure water at this temperature is 35 mm Hg, lowering of vapour pressure will be.
(Molar mass of urea = 60g mol^{-1})

(1) 0.027 mmHg (2) 0.031 mmHg

(3) 0.017 mmHg (4) 0.028 mmHg

CSO226

34. Molal depression constant for a solvent is 4.0 K kg mol^{-1}. The depression in the freezing of the solvent for 0.03 mol kg^{-1} solution of K_2SO_4 is (Assume complete dissociation of the electrolyte)

(1) 0.18 K (2) 0.36 K

(3) 0.12 K (4) 0.24 K

CSO227

35. K_2HgI_4 is 40% ionised in aqueous solution. The value of its van't Hoff factor (i) is

(1) 1.6 (2) 1.8 (3) 2.2 (4) 2.0

CSO228

36. Elevation in the boiling point for 1 molal solution of glucose is 2 K. The depression in the freezing point for 2 molal solution of glucose in the same solvent is 2 K. The relation between K_b and K_f is

(1) $K_b = 1.5\ K_f$ (2) $K_b = 0.5\ K_f$

(3) $K_b = K_f$ (4) $K_b = 2K_f$

CSO229

37. A solution contain 62 g of ethylene glycol in 250 g water is cooled upto −10°C. If K_f for water is 1.86 K kg mol^{-1}, then amount of water (in g) separated as ice is

(1) 32 (2) 48 (3) 64 (4) 16

CSO230

38. The freezing point (in°C) of solution containing 0.1 g of $K_3[Fe(CN)_6]$ (mol. wt 329) in 100 g of water (K_f = 1.86 k kg mol^{-1}) is

(1) -2.3×10^{-2} (2) -5.7×10^{-2}

(3) -5.7×10^{-3} (4) -1.2×10^{-2}

CSO231

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**2**	**1**	**1**	**4**	**1**	**2**	**3**	**2**	**3**	**1**	**2**	**1**	**4**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**3**	**2**	**2**	**3**	**3**	**4**	**4**	**1**	**3**	**2**	**4**	**2**	**1**	**1**
Question	31	32	33	34	35	36	37	38							
Answer	**3**	**3**	**3**	**2**	**2**	**4**	**3**	**1**							

Chemical Kinetics

1. In gaseous reaction it is important for the understanding of the upper atmosphere H_2O and O react bimolecularly to form two OH radicals. ΔH for this reaction is 72 kJ at 500 K and E_a is 77 kJ, then E_a for the bimolecular recombination of two OH radicals to form H_2O and O is :-

(1) 3 kJ (2) 4 kJ

(3) 5 kJ (4) 7 kJ

CCK185

2. Half lives and initial concentration of a first order and zero order reactions are same. Then the ratio of the initial rates of the first order reaction to that of zero order reaction is :-

(1) 1/0.693 (2) 2 × 0.693

(3) 2/0.693 (4) 6.93

CCK186

3. In the following first order reactions

(A) $\xrightarrow{K_1}$ Product, (B) $\xrightarrow{K_2}$ Product,

the ratio K_1/K_2 if 90% of (A) has been reacted in time 't' while 99% of (B) has been reacted in time 2t is :-

(1) 1

(2) 2

(3) 1/2

(4) $2\left(\frac{1-2\log 3}{2-\log 11-2\log 3}\right)$

CCK187

4. For a reaction : A $\longrightarrow$ Product

Rate law is $-\frac{d[A]}{dt} = K[A]_0$

The concentration of A left after time t when t = 1/K is :-

(1) $\frac{[A]_0}{e}$ (2) $[A]_0 e$

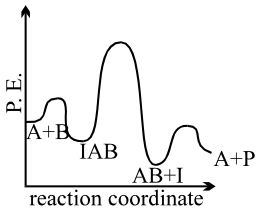
(3) $\frac{[A]_0}{e^2}$ (4) $\frac{1}{[A]_0}$

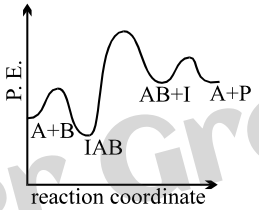
CCK188

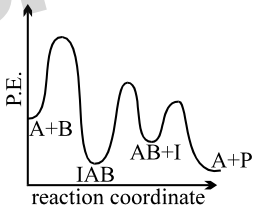
5. The following mechanism has been proposed for the exothermic catalyzed complex reaction.

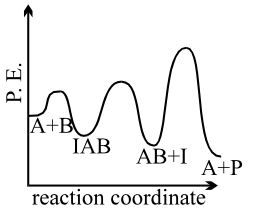
$A + B \underset{}{\overset{Fast}{\rightleftharpoons}} IAB \xrightarrow{k_1} AB + I \xrightarrow{k_2} P + A$

If k_1 is much smaller than k_2. The most suitable qualitative plot of potential energy (P.E.) versus reaction coordinate for the above reaction.

(1) 

(2) 

(3) 

(4) 

CCK189

6. The reaction A(g) + 2B(g) $\longrightarrow$ C(g) + D(g) is an elementary process. In an experiment, the initial partial pressure of A & B are P_A = 0.6 and P_B = 0.8 atm. Calculate the ratio of rate of reaction relative to initial rate when P_C becomes 0.2 atm.

(1) $\frac{1}{4}$ (2) $\frac{1}{10}$

(3) $\frac{1}{6}$ (4) 2

CCK190

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7. In the following reaction, rate constant is 1.2×10^{-2} M s^{-1} A $\longrightarrow$ B. What is concentration of B after 10 min., if we start with 10 M of A.

(1) 2.8 M (2) 7.2 M
(3) 8.2 M (4) 2.7 M

CCK191

8. Two substances A ($t_{1/2}$= 5 min) and B ($t_{1/2}$ = 15 min) follow first order kinetics are taken in such a way that initially [A]= 4[B]. Calculate the time after which the concentration of both the substance will be equal.

(1) 15 min (2) 20 min
(3) 10 min (4) 25 min

CCK192

9. A reaction takes place in three steps with individual rate constant and activation energy as follows-

	Rate constant	Activation energy
Step 1	k_1	$E_{a_1} = 180 kJ/mol$
Step 2	k_2	$E_{a_2} = 80 kJ/mol$
Step 3	k_3	$E_{a_3} = 50 kJ/mol$

If overall rate constant, (k) = $\left(\frac{k_1 k_2}{k_3}\right)^{2/3}$, then overall activation energy of the reaction will be:

(1) 140 kJ/mol (2) 150 kJ/mol
(3) 130 kJ/mol (4) 120 kJ/mol

CCK193

10. A first order reaction is 50% completed in 20 minutes at 27°C and in 5 min at 47°C. The energy of activation of the reaction is

(1) 43.85 kJ/mol (2) 55.14 kJ/mol
(3) 11.97 kJ/mol (4) 6.65 kJ/mol

CCK194

11. Under the same reaction conditions, initial concentration is 1.386 mol dm^{-3} of a substance becomes half in 40 seconds and 20 seconds through first order and zero order kinetics, respectively. Ratio $\left(\frac{k_1}{k_0}\right)$ of the rate constants for first order (k_1) and zero order (k_0) of the reactions is

(1) 0.5 mol^{-1} dm^3 (2) 1.0 mol dm^{-3}
(3) 2.0 mol^{-1} dm^3 (4) 0.1 mol dm^{-3}

CCK195

12. The rate constant of a first order reaction at 37°C is 10^{-3} min^{-1}. The temperature coefficient of this reaction is 2. What is the rate constant (in min^{-1}) at 17°C for this reaction ?

(1) 10^{-3} (2) 5×10^{-4}
(3) 2.5×10^{-3} (4) 2.5×10^{-4}

CCK196

13. Consider a chemical reaction involving compounds A and B, which is found to be first order in A and second order in B. At what rate will the reaction occur in experiment 2 ?

Experiment	Rate (Ms^{-1})	Initial [A]	Initial [B]
1	0.10	1.0 M	0.20 M
2	?	2.0 M	0.60 M

(1) 1.8 M s^{-1} (2) 0.20 M s^{-1}
(3) 1.2 M s^{-1} (4) 0.36 M s^{-1}

CCK197

14. In a reaction carried out at 400 K, 0.0001% of the total number of molecules are in activated state. The energy of activation of the reaction is.

(1) zero
(2) 7.37 k cal/mol
(3) 9.212 k cal/mol
(4) 11.05 k cal/mol

CCK198

15. A solution of N_2O_5 in CCl_4 yields 5 mL of O_2 by decomposition after 20 minutes from the start of the experiment and 10 mL of O_2 after a very long time. The decomposition obeys first order kinetics. What volume of O_2 would have evolved, 40 minutes after the start

(1) 7.5 mL (2) 7.25 mL
(3) 1.5 mL (4) 6 mL

CCK199

16. Two first-order reactions have half-lives in the ratio 3 : 2. Calculate the ratio of time intervals
$t_1 : t_2$, t_1 is the time period for 25% completion of the first reaction and t_2 for 75% completion of the second reaction :-

(1) 0.311 : 1 (2) 0.420 : 1
(3) 0.273 : 1 (4) 0.119 : 1

CCK200

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17. A certain reaction obeys the rate equation (in the integrated form) $\left[C^{(1-n)} - C_0^{(1-n)}\right] = (n-1)\,kt$ where n is the order of the reaction; C_0 the initial concentration and C, the concentration after time t. What is the unit of k for n = 3 :-

(1) sec^{-1}

(2) litre mol^{-1} sec^{-1}

(3) mol $litre^{-1}$ sec^{-1}

(4) $litre^2$ mol^{-2} sec^{-1}

CCK201

18. The half-life of a first order reaction is one hour. During what time interval will the concentration be reduced to $\frac{7}{8}$ of its initial value :-

(1) 6.32 minutes (2) 17.4 minutes

(3) 23.2 minutes (4) 11.6 minutes

CCK202

19. The time elapsed between 33% and 67% completion of a first order reaction is 30 minutes What is the time needed for 25% completion :-

(1) 15.5 min (2) 12.3 min

(3) 18.5 min (4) 16.5 min

CCK203

20. Kinetics of the reaction, A(g) $\longrightarrow$ 2B(g) + C(g) is followed by measuring the total pressure at different times. It is given that :

Initial pressure of A = 0.5 atm.

Total pressure of reaction mixture after 2 hours = 0.7 atm.

Rate constant of the reaction = 1×10^{-3} s^{-1}

What is the rate of reaction when the total pressure is 0.7 atm ?

(1) 2.0×10^{-4} atm s^{-1} (2) 4.0×10^{-4} atm s^{-1}

(3) 5.0×10^{-4} atm s^{-1} (4) 7.0×10^{-4} atm s^{-1}

CCK204

21. For $N_2 + 3H_2 \rightarrow 2NH_3 + 22$ kcal, E_a (energy of activation) is 70 kcal. Hence, E_a of $2NH_3 \rightarrow N_2 + 3H_2$ is :-

(1) 92 kcal (2) 70 kcal

(3) – 70 kcal (4) – 92 kcal

CCK205

22. A gaseous reaction, $2A(g) \rightarrow B(g) + 5C(g)$ shows increase in pressure from 80 mm to 100 mm in 5 minutes. The rate of disappearance of A is :-

(1) 4 mm min^{-1} (2) 8 mm min^{-1}

(3) 16 mm min^{-1} (4) 2 mm min^{-1}

CCK206

23. On introducing a catalyst at 500K, the rate constant of a first order reaction increases 2.718 times. If the activation energy in the presence of a catalyst is 4.15 kJ mol^{-1}. Then what will be E_a in absence of catalyst (value of e = 2.718):-

(1) 4.15 kJ (2) 2.08 kJ

(3) 2.718 kJ (4) 8.3 kJ

CCK207

24. For a first order reaction A $\rightarrow$ products, the concentration of [A] is reduced from 2M to 0.125 M in one hour, the $t_{1/2}$ of this reaction (in min) is :-

(1) 30 (2) 45 (3) 15 (4) None

CCK208

25. In the following reaction; $xA \longrightarrow yB$

$$\log_{10}\left[\frac{d[A]}{dt}\right] = \log_{10}\left[\frac{d[B]}{dt}\right] + 0.3010$$

A and B respectively can be

(1) n–butane and iso-butane

(2) C_2H_2 and C_6H_6

(3) C_2H_4 and C_4H_8

(4) N_2O_4 and NO_2

CCK209

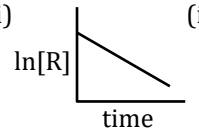
26. For the reaction of H_2 with I_2, the rate constant is 2.5×10^{-4} dm^3 $mol^{-1}s^{-1}$ at 327°C and 1.0 dm^3 $mol^{-1}s^{-1}$ at 527°C. The activation energy for the reaction, in kJ mol^{-1} is (R=8.314 JK^{-1} mol^{-1})?

(1) 59 (2) 72 (3) 150 (4) 166

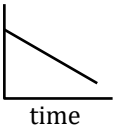
CCK210

27. The given plots represent the variation of the concentration of a reaction R with time for two different reactions (i) and (ii). The respective orders of the reactions are.

(i)



(ii)



(1) 1, 1 (2) 0, 2 (3) 0, 1 (4) 1, 0

CCK211

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28. For the reaction, $2A + B \longrightarrow C$, the values of initial rate at different reactant concentrations are given in the table below. The rate law for the reaction is.

[A] (mol L^{-1})	[B] (mol L^{-1})	Initial rate (mol $L^{-1}s^{-1}$)
0.05	0.05	0.045
0.10	0.05	0.090
0.20	0.10	0.72

(1) rate = $k[A][B]^2$

(2) rate = $k[A]^2[B]^2$

(3) rate = $k[A][B]$

(4) rate = $k[A]^2[B]$

CCK212

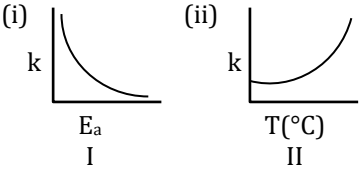
29. The reaction, $2X \rightarrow B$ is a zeroth order reaction. If the initial concentration of X is 0.2 M, the half-life is 6 hr. When the initial concentration of X is 0.5 M, the time required to reach its final concentration of 0.2 M will be

(1) 7.2 h (2) 18.0 h

(3) 12.0 h (4) 9.0 h

CCK213

30. Consider the given plots for a reaction obeying Arrhenius equation ($0°C < T < 300°C$): (k and E_0 are rate constant and activation energy, respectively)



Choose the correct option.

(1) Both I and II are wrong

(2) Both I and II are correct

(3) I is wrong but II is right

(4) I is right but II is wrong

CCK214

31. The following results were obtained during kinetic studies of the reaction;

$2A + B \longrightarrow$ Products

Experiment	[A] (in mol L^{-1})	[B] (in mol L^{-1})	Initial rate of reaction (in mol $L^{-1}min^{-1}$)
I.	0.10	0.20	6.93×10^{-3}
II.	0.10	0.25	6.93×10^{-3}
III.	0.20	0.30	1.386×10^{-2}

The time (in minutes) required to consume half of A is

(1) 5 (2) 10 (3) 100 (4) 1

CCK215

32. Higher order (>3) reactions are rare due to

(1) low probability of simultaneous collision of all the reacting species

(2) increase in entropy and activation energy as more molecules are involved

(3) shifting of equilibrium towards reactants due to elastic collisions

(4) loss of active species on collision

CCK216

33. For the non-stoichiometric reaction, $2A + B \rightarrow C + D$, the following kinetic data were obtained in three separate experiments, all at 298K

	Initial concentration [A]	Initial concentration [B]	Initial rate of formation of C (mol $L^{-1}s^{-1}$)
I.	0.1 M	0.1 M	1.2×10^{-3}
II.	0.1 M	0.2 M	1.2×10^{-3}
III.	0.2 M	0.1 M	2.4×10^{-3}

The rate law for the formation of C is

(1) $\frac{dC}{dt} = k[A][B]$ (2) $\frac{dC}{dt} = k[A]^2[B]$

(3) $\frac{dC}{dt} = k[A][B]^2$ (4) $\frac{dC}{dt} = k[A]$

CCK217

34. The rate of a reaction doubles when its temperature changes from 300K to 310K. Activation energy of such a reaction will be ($R = 8.314\ JK^{-1}\ mol^{-1}$ and $\log 2 = 0.301$)

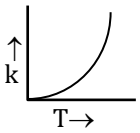
(1) 53.6 KJ mol^{-1} (2) 48.6 KJ mol^{-1}

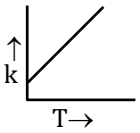
(3) 58.6 KJ mol^{-1} (4) 60.5 KJ mol^{-1}

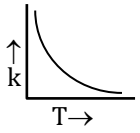
CCK218

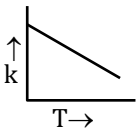
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35. Plots showing the variation of the rate constant (k) with temperature (T) are given below. The plot that follows Arrhenius equations is.

(1) 

(2) 

(3) 

(4) 

CCK219

36. For a first order reaction, A $\rightarrow$ P, the temperature (T) dependent are constant (k) was found to follow the equation:

$$\log k = \frac{2000}{T} + 6.0$$

the pre-exponential factor A and the activation energy E_a, respectively, are

(1) $1.0 \times 10^6\ s^{-1}$ and 9.2 KJ mol^{-1}

(2) $6.0\ s^{-1}$ and 16.6 KJ mol^{-1}

(3) $1.0 \times 10^6\ s^{-1}$ and 16.6 KJ mol^{-1}

(4) $1.0 \times 10^6\ s^{-1}$ and –38.3 KJ mol^{-1}

CCK220

37. Which one of the following statement(s) is incorrect about order of reaction ?

(1) Order of reaction is determined experimentally

(2) Order of reaction is equal to sum of the power of concentration terms in differential rate law

(3) It is not affected with stoichiometric coefficient of the reactants

(4) Order cannot be fractional

CCK221

38. Consider the chemical reaction,

$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

The rate of this reaction can be expressed in terms of time derivatives of concentration of $N_2(g)$, $H_2(g)$ or $NH_3(g)$, Identify the correct relationship amongst the rate expressions.

(1) $\text{Rate} = -\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

(2) $\text{Rate} = -\frac{d[N_2]}{dt} = -3\frac{d[H_2]}{dt} = 2\frac{d[NH_3]}{dt}$

(3) $\text{Rate} = \frac{d[N_2]}{dt} = \frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$

(4) $\text{Rate} = -\frac{d[N_2]}{dt} = -\frac{d[H_2]}{dt} = \frac{d[NH_3]}{dt}$

CCK222

39. A catalyst is a substance which

(1) increases the equilibrium concentration of the product

(2) changes the equilibrium constant of the reaction

(3) shortens the time to reach equilibrium

(4) supplies energy to the reaction

CCK223

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**2**	**1**	**1**	**1**	**3**	**2**	**1**	**1**	**2**	**1**	**4**	**1**	**4**	**1**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**1**	**4**	**4**	**2**	**2**	**1**	**4**	**4**	**3**	**3**	**4**	**4**	**1**	**2**	**2**
Question	31	32	33	34	35	36	37	38	39						
Answer	**2**	**1**	**4**	**1**	**1**	**4**	**4**	**1**	**3**						